Otter Weather: Skillful and Computationally Efficient Medium-Range Weather Forecasting

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Abstract

State-of-the-art medium-range AI weather models can outperform traditional Numerical Weather Prediction (NWP) but require massive training budgets. This restricts usage for under-resourced groups and severely limits fast model iteration. Here we develop Otter Weather, a highly efficient spatiotemporal forecasting model designed to democratise high-performance weather prediction with AI. Evaluated on ERA5 reanalysis data at 1.5° resolution using standard WeatherBench protocols, the Otter family significantly advances the skill-compute Pareto frontier. The deterministic version outperforms the best NWP baseline by 9.6% at a 24-hour lead time while requiring fewer than 3.5 A100-days for training. It provides a 2× efficiency gain over lightweight AI models and a 100-fold reduction in compute compared to resource-intensive frontier architectures. We extend these efficiency gains into probabilistic forecasting by training via the Continuous Ranked Probability Score (CRPS). Scaling to a larger architecture, Otter-XL achieves a 9.7% CRPS improvement over the IFS ENS baseline. This yields an almost two-fold increase in predictive skill over comparable lightweight models at similar compute budgets. Otter-XL also outperforms frontier architectures like GenCast by over 2%, while using an order of magnitude less compute. Finally, Otter is applied out-of-the-box to a complex acoustic scattering PDE task where it outperforms a state-of-the-art foundation modelling approach, suggesting that the advances made here might apply across a range of scientific domains.

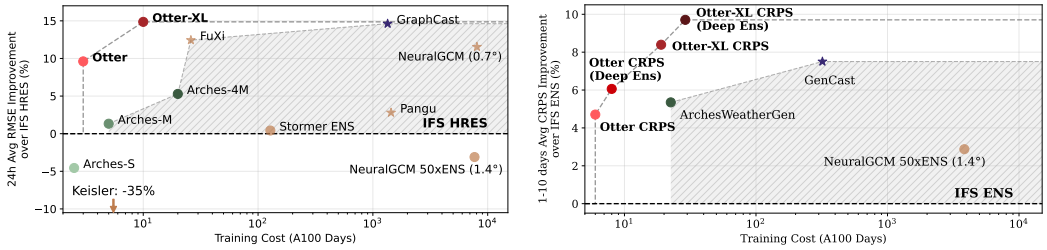
1 Introduction

The field of global weather forecasting is being transformed by data-driven approaches. AI models have reached performance levels comparable to strong numerical weather prediction (NWP) baselines, even exceeding them in many tasks [Lam et al., 2023, Lang et al., 2024, Alet et al., 2025, Bodnar et al., 2025]. This paradigm shift holds the potential to democratise weather forecasting, but fully realising this vision now rests on reducing training costs so that groups outside of well-resourced AI laboratories can design, train and control their own models.

Lowering the training cost barrier would empower a diverse range of beneficiaries—from academic institutions and under-resourced operational agencies to startups—to advance the field in two comple-

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(a) Deterministic Models

(b) Probabilistic Models

Figure 1: **Skill score over IFS HRES/ENS on the headline variables.** (a) RMSE at 24h. (b) CRPS averaged over 1-10 days. Circles (○) represent models trained at lower resolutions (1°/1.4°/1.5°), while stars (⋆) indicate higher-resolution training (0.25°/0.7°). The models in the **Otter Weather** family advance both the deterministic and probabilistic the Pareto frontiers of skill versus compute.

mentary ways. First, it enables such groups to train competitive weather models from scratch rather than being restricted to fine-tuning released foundation models. This pre-training capability is essential for scientific independence: it lets researchers rigorously audit, reproduce, and extend models without relying on the original developers, and to tailor architectures to specific regions, variables, or downstream tasks. Second, regardless of available resources, a rapid iteration cycle accelerates model development, retraining and deployment for all practitioners, and speeds up scientific discovery.

At present, however, the field remains far from this level of accessibility. While the community is becoming less reliant on the CPU-based supercomputers required for traditional NWP, current state-of-the-art (SOTA) data-driven methods [Lam et al., 2023, Price et al., 2025, Alet et al., 2025] have established a new barrier to entry: they require massive, distributed clusters of hundreds of GPUs or TPUs—infrastructure accessible only to well-resourced AI laboratories. In this work, we investigate whether this paradigm can be shifted to make high-performance weather forecasting accessible to practitioners with limited resources, down to a single GPU. To do so, we first identify the primary drivers of current computational costs. Many SOTA architectures rely on heavy inductive biases, such as custom graph neural networks (GNNs) [Lam et al., 2023, Price et al., 2025] or complex spherical geometry operators [Mahesh et al., 2025]. While physically motivated, these specialised components often constrain the model and lack the hardware optimisation of mainstream techniques used in Large Language Models (LLMs) and computer vision.

To date, the most efficient competitive models are ArchesWeather [Couairon et al., 2024], ERDM [Cachay et al., 2025], and U-Cast [Cachay et al., 2026]. While the deterministic ArchesWeather model requires 5 A100-days of training, probabilistic models like ArchesWeatherGen and U-Cast demand over 20, and ERDM exceeds 40. In this work, we push this efficiency further by prioritising general-purpose, highly optimised methods over bespoke, domain-specific complexity.

This strategy yields a dual advantage: it utilises architectures proven to scale across diverse domains, while simultaneously exploiting the well-developed hardware support and low-level optimisations of mainstream AI development. Specifically, we challenge the prevailing assumption that Earth-specific geometric priors are prerequisites for SOTA performance; instead, we use a standard 2D Swin Transformer [Liu et al., 2021] augmented with modern advances from language modelling, including Rotary Positional Embeddings (RoPE) [Su et al., 2024] and SwiGLU activations [Shazeer, 2020], all trained via the efficient Muon optimiser [Jordan et al., 2024]. By leveraging these established foundations, we advance both the deterministic and probabilistic skill-compute Pareto frontiers, setting a powerful yet accessible standard for spatiotemporal modelling.

Our core contributions are:

1. **Democratising Deterministic Weather.** We introduce **Otter**, a 2D Swin-UNet Transformer model that establishes a new skill-compute Pareto frontier (fig. 1a). Trained on a single GPU in under 3.5 A100 days, Otter outperforms leading low-compute AI forecasting models. Furthermore, our scaled model, **Otter-XL**, surpasses FuXi's [Chen et al., 2023] performance at ∼3x lower compute cost and achieves skill at 1.5° resolution comparable to GraphCast [Lam et al., 2023] while requiring ∼130x less training compute.

2. **Low-Compute Probabilistic Forecasting.** We demonstrate that our architectural advances prove equally effective in the probabilistic setting (fig. 1b), allowing us to significantly advance the probabilistic skill-compute Pareto frontier. At the base scale, Otter yields competitive forecasts with a total compute budget under 8 A100-days — less than half the $>$ 20 A100-day budget of recent efficient models-outperforming ArchesWeatherGen and remaining highly competitive with U-Cast. Otter-XL establishes a new standard: requiring approximately 30 A100-days, it delivers an almost two-fold skill increase over comparable lightweight models, and surpasses resource-intensive frontier architectures like GenCast by over 2% while using an order of magnitude less compute.
3. **Toward Generality Across Dynamical Systems.** Since Otter is built from domain-agnostic components, we hypothesise it may transfer broadly to spatiotemporal modelling problems. To test this cross-domain hypothesis, we evaluate our approach on an acoustic scattering partial differential equation (PDE) task from the Well benchmark [Ohana et al., 2024]. Otter achieves strong performance out-of-the-box against foundation models like Walrus [McCabe et al., 2026], suggesting a promising path toward efficient models across the physical sciences.
4. **Reevaluating Inductive Biases & A Practical Guide.** We provide a systematic ablation study demonstrating that standard, hardware-optimised components (RoPE, SwiGLU, Muon) are sufficient for high-skill forecasting. We distill these findings into a practical training recipe for practitioners under resource constraints, while also documenting explored techniques whose marginal gains did not justify their implementation overhead.

2 Related Work

2.1 Data-Driven Global Weather Forecasting

Specialised and Foundation Models. Recent breakthroughs in data-driven weather forecasting have seen learned models match or surpass numerical weather prediction (NWP) baselines. However, models achieving state-of-the-art (SOTA) performance generally rely on heavy, domain-specific architectural modifications. For instance, Pangu-Weather [Bi et al., 2023] uses 3D Earth-specific Transformers, while other models employ complex Graph Neural Networks (GNNs) [Keisler, 2022, Lam et al., 2023, Price et al., 2025, Lang et al., 2024, Alet et al., 2025] or spectral approaches like Spherical Fourier Neural Operators [Pathak et al., 2022, Bonev et al., 2025]. These physical priors often incur massive training overheads—hundreds of GPU-days—and require custom-written kernels [Fu et al., 2024]. Similarly, adapting foundation models like ClimaX [Nguyen et al., 2023] and Aurora [Bodnar et al., 2025] necessitates complex engineering and highly restrictive hardware requirements [Subich, 2025, Lehmann et al., 2025], establishing a severe barrier to democratisation. Beyond these hardware constraints, such a reliance on pre-trained models inherently tethers researchers to the architectural choices, variable selections, and latent biases of the original developers. This dependency restricts the scientific sovereignty required to tailor models to independent research objectives or specific regional needs, even when fine-tuning is computationally feasible.

Efficient Architectures and Probabilistic Modelling. In response, a counter-trend has emerged focusing on compute-efficient architectures that minimise inductive biases, such as Stormer [Nguyen et al., 2024], ArchesWeather [Couairon et al., 2024], and the concurrent work U-Cast [Cachay et al., 2026]. Simultaneously, the field is shifting toward probabilistic modelling to capture predictive uncertainty. While flow-based and diffusion models (e.g., GenCast [Price et al., 2025], ArchesWeatherGen [Couairon et al., 2024]) achieve this, they require computationally expensive iterative denoising at inference time. Alternatively, optimising proper scoring rules like the Continuous Ranked Probability Score (CRPS)—a technique pioneered by FGN [Alet et al., 2025] and AIFS-CRPS [Lang et al., 2026], and inspired by PDE literature [Diaconu et al., 2026]—offers a highly efficient route to fine-tune a deterministic checkpoint into a probabilistic model. U-Cast [Cachay et al., 2026] recently applied this concept, using Monte Carlo (MC) dropout in a U-Net architecture to build a deep ensemble from multiple CRPS fine-tuned checkpoints. However, models like U-Cast and ArchesWeatherGen still operate at training budgets exceeding 20 A100-days. In contrast, our work pushes the Pareto frontier even further by leveraging CRPS training with MC dropout within a strictly lower-compute, transformer-based setting. Rather than undertaking independent fine-tuning runs to construct an ensemble, we use an efficient methodology that reuses checkpoints already generated during standard hyperparameter tuning, achieving competitive probabilistic performance at a fraction of the computational cost.

2.2 ML Breakthroughs from Vision and Language Modelling

As weather models converge with general-purpose architectures, the broader transferability of modern techniques from the LLM and computer vision communities remains under-explored. While ArchesWeather successfully integrated SwiGLU activations [Shazeer, 2020], we argue the field still lacks a systematic evaluation of domain-agnostic adaptations. We ablate SOTA mechanisms, including Rotary Positional Embeddings (RoPE) [Su et al., 2024], the Muon optimiser [Jordan et al., 2024], Mixture of Experts [Shazeer et al., 2017], Hyper-Connections [Zhu et al., 2025], Neighborhood Attention [Hassani et al., 2023], and Masked Autoencoder pretraining [He et al., 2022].

Positioning Otter. Against this landscape, our approach distinguishes itself by: (1) proposing a highly optimised, ablation-driven deterministic foundation that strips away unnecessary domain complexity; (2) reusing this architecture for a computationally cheap probabilistic treatment (either through fine-tuning or training from scratch), halving the training cost of efficient baselines while entirely avoiding slow iterative inference; and (3) demonstrating cross-domain generality by successfully applying this exact architecture to an acoustic scattering PDE task.

3 Experimental Setup and Architectural Foundation

3.1 Task Specification

Following the well-established methodology for assessing medium-range AI weather models [Couairon et al., 2024, Nguyen et al., 2024, Lam et al., 2023], the task is to predict the temporal evolution of atmospheric variables using the ERA5 reanalysis dataset [Hersbach et al., 2020]. We predict future atmospheric states $X_{t+\delta t}$ at a lead time of $\delta t = 6$ hours, conditioned on a recent history of atmospheric states. The model predicts the residual difference relative to the most recent observation, and longer horizons are generated via autoregressive rollout. The state vector X_t comprises 87 variables: $AV = 6$ atmospheric variables X_t^{atm} (temperature, geopotential, specific humidity, vertical velocity, and U/V wind components) across 13 pressure levels, $SV = 4$ surface variables X_t^{surf} (2m temperature, mean sea-level pressure, and 10m U/V wind components), and 5 static topographic features. We train on data from 1979 to 2019 and evaluate on 2020, with forecasts initialised twice daily at 00:00 and 12:00 UTC. We provide additional architecture and training details in appendix B.

Spatial resolution. We conduct all investigations at a spatial resolution of 1.5°. While some SOTA models operate at 0.25°, prior work [Allen et al., 2025] establishes that 1.5° is a computationally efficient and reliable proxy for ablating architectural decisions, especially in the deterministic setting, where forecasting skill (i.e., RMSE) is primarily derived from low spatial frequency components. Furthermore, 1.5° remains a widely adopted evaluation standard even for high-resolution models [Rasp et al., 2024], with skill at 1.5° being highly predictive of performance at 0.25°.

Evaluation Protocol. Following the standard WeatherBench 2 evaluation protocol [Rasp et al., 2024] at 1.5°, we adopt established metrics and horizons suited to the respective modelling paradigms, separating the evaluation into deterministic and probabilistic. Furthermore, we distinguish between models evaluated with and without rollout fine-tuning (RFT)—a technique used to align the training objective with test-time behaviour, mitigating the train-test discrepancy for models that are initially trained with a one-step loss but evaluated via autoregressive rollouts. All metrics are averaged over the headline variables (Z500, Q700, T850, U850, V850, T2m, SP, U10m, and V10m) and over increments of 24h for a fair comparison to ArchesWeather which uses a 24h lead time.

- **Deterministic Evaluation:** We measure performance using the latitude-weighted Root Mean Squared Error (RMSE), restricting our focus to lead times up to 72 hours. This time restriction is necessary because models optimised via Mean Squared Error (MSE) target the conditional mean; at extended lead times, the considerable predictive uncertainty causes this averaging effect to produce unphysically smooth, blurred fields [Ben Bouallègue and the AIFS team, 2023]. We divide this evaluation into two settings:
  - *Ablations (non-RFT):* To isolate architectural and optimisation choices from the confounding effects of RFT, we evaluate non-RFT models in our ablations. We report the average RMSE for 6–24h lead times—a window long enough to draw robust conclusions, yet short enough to avoid rewarding oversmoothing.

– *Comparison to Baselines (RFT):* Because external SOTA baselines use different internal time-steps (e.g., 12 or 24 hours), we cannot apply the averaged evaluation protocol described above. For a fair comparison among autoregressive models, we uniformly align evaluation at a 24-hour lead time. We report the aggregated skill of our *RFT* models here, with extended evaluations up to 3 days provided in the Appendix.

- **Probabilistic Evaluation:** Because probabilistic models sample from a learned distribution rather than a smoothed mean, they enable meaningful evaluation at longer horizons. To obtain **Otter CRPS** and **Otter-XL CRPS**, the probabilistic models within the Otter family, we retrofit the deterministic backbone using a CRPS objective, followed by RFT. Performance is measured via the Continuous Ranked Probability Score (CRPS) applied to the rolled out model predictions up to 10 days. These CRPS scores are then averaged to provide a single headline number. Additionally, we evaluate a probabilistic deep ensemble constructed by applying diverse RFT hyperparameter configurations to the same base probabilistic checkpoint (**Otter(-XL) CRPS (Deep Ens)**).

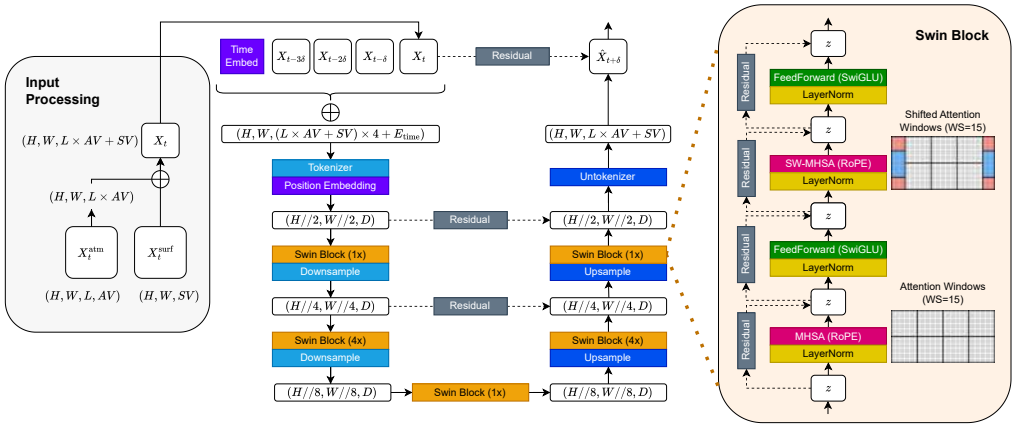


Figure 2: **Otter Weather architecture.** A 2D Swin Transformer predicts the future state $X_{t+\delta}$ from recent history $X_{t-(L-1)\times\delta}, \ldots, X_t$. Atmospheric variables across 13 pressure levels, surface variables, and time embeddings (E_{time}) are concatenated channel-wise at a spatial resolution of $H \times W = 121 \times 240$. Within the shifted-window attention (SW-MHSA) blocks, we employ RoPE and omit attention masks to enable cross-boundary communication, indicated by the shaded regions.

3.2 The Otter Base Architecture

Our design strictly minimises Earth-specific modifications, favouring standard, widely supported architectural components. This lets us directly use mature, low-level hardware optimisations and integrate architectural advancements proven across other AI domains, thereby maximising training efficiency. Furthermore, avoiding bespoke architectures lets the model be retrained on arbitrary variables without redesigning explicit pressure-level or variable-specific encoders.

Input Representation. Unlike architectures employing separate encoders for surface and upper-atmosphere variables [Lam et al., 2023, Couairon et al., 2024], we simply concatenate all input variables along the channel dimension (fig. 2). This tensor comprises SV surface variables, AV atmospheric variables, and 32 continuous multi-scale Fourier time embeddings. It is tokenised using a strided convolutional layer with patch size P and token dimensionality D.

Backbone. We use a 2D Swin Transformer [Liu et al., 2021] arranged in a symmetric U-Net configuration (Encoder, Bottleneck, Decoder). We deviate from the standard Swin in three ways: (1) we maintain a constant token dimensionality D across the U-Net; (2) we replace standard GeLU MLPs with SwiGLU feed-forward networks [Shazeer, 2020]; and (3) we replace relative position biases with 2D Rotary Positional Embeddings (RoPE) [Su et al., 2024]. Our Base configuration uses $[2, 8, 4]$ blocks per stage, 16 attention heads, and $D = 1536$.

Attention Masking and Topology. Standard models often apply custom attention masks to approximate a cylinder topology (masking the poles). This incurs significant compute overhead and breaks

optimised attention kernels. Instead, we remove masking entirely, enabling communication across all boundaries (a torus topology) and relying on the positional embeddings to learn the spatial structure.

Optimisation and Regularisation. Otter is trained to minimise latitude- and pressure-weighted RMSE using the Muon optimiser [Jordan et al., 2024], which orthogonalises parameter updates to accelerate convergence. To offset Muon's matrix-operation overhead at small batch sizes, we use gradient accumulation. We apply strong regularisation typical of LLMs, including weight decay (WD $= 0.15$), dropout ($p = 0.1$), and a batch-wise stochastic depth strategy (drop path probability 0.1), which skips residual blocks entirely for a mini-batch, improving both stability and training speed.

4 Pushing the Pareto Frontier in Weather Forecasting

4.1 Optimising the Deterministic Backbone

A critical advantage of Otter Weather's minimal hardware requirements is the rapid feedback loop: with training completing in under 3.5 A100-days, we can democratise architectural search. We systematically ablate key components to isolate high-impact design choices, evaluating their computational trade-offs averaged over 6-24h lead times (without RFT) to isolate architectural capacity.

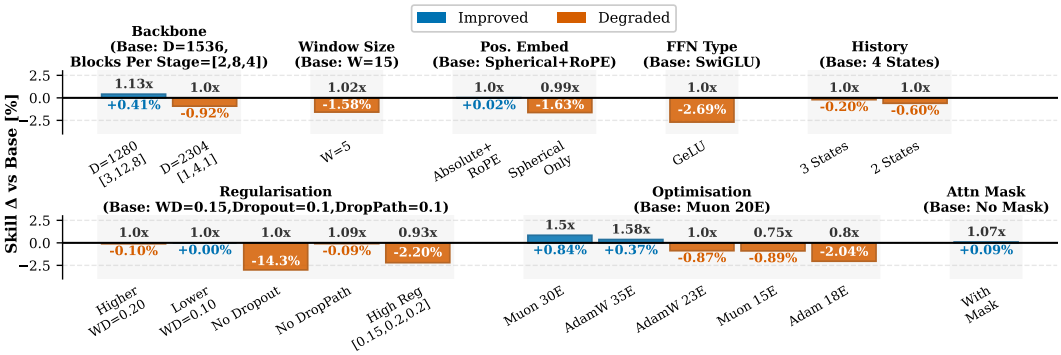


Figure 3: **Architectural and training ablations.** Bars show the relative percentage change in predictive skill (6–24h RMSE average of non-RFT models) versus the base model (black zero-line). Blue indicates improvement; orange denotes degradation. Multipliers above each bar reflect the relative computational cost. Deviations from the base configuration either degrade performance or incur disproportionate computational overheads that outweigh any marginal gains.

Ablation Insights. Our ablations (fig. 3) challenge several prevailing assumptions in weather modelling. Regarding the ***backbone*** profile, we found that a balanced, deeper U-Net configuration—distributing compute evenly across stages (e.g., $[2, 8, 4]$ blocks with $D = 1536$)— outperformed a "wide but shallow" setup ($[1, 4, 1]$ blocks with $D = 2304$) at a matched compute budget. We also found that optimising the ***FFN Type*** (i.e., replacing GeLU activations with SwiGLU) and the ***Pos. Embed*** (i.e., utilising 2D Rotary Positional Embeddings (RoPE)) yield consistent skill improvements. Surprisingly, omitting explicit attention masking (***Attn Mask***) across the Earth's poles (treating the topology as a torus rather than a cylinder) incurred negligible performance drops while drastically simplifying the implementation and providing a slight computational speedup. For ***optimisation***, the Muon optimiser consistently outperformed compute-matched AdamW across all evaluated training durations. Notably, the performance gap between the two widened as the number of epochs decreased, making Muon exceptionally advantageous for compute-constrained, rapid-iteration regimes. Effective ***regularisation*** proved vital, with dropout ($p = 0.10$) and strong weight decay (WD $= 0.15$) yielding the largest gains.

Evaluated and Excluded Components. Consistent with our goal of architectural simplicity and efficiency, we discarded several popular techniques from vision and language—including Hyper-connections, Neighborhood Attention (NATTEN), Masked Autoencoder (MAE) pre-training, and Mixture of Experts (MoE). While these are prominent in other domains, we found their marginal performance gains did not justify the added engineering complexity, dependency on custom kernels (in the case of NATTEN), or training fragility (for MoE). By documenting these ineffective directions,

we aim to provide a transparent account of which modern advancements transfer readily to weather forecasting and which require significant tuning or customisation.

Deterministic Performance and Scaling. By combining these optimal, hardware-friendly components, the resulting **Otter Weather** base model extends the deterministic Pareto frontier (fig. 1a). After RFT, Otter achieves a **9.6% improvement** over the operational IFS HRES baseline across headline variables. Crucially, it surpasses the computationally heavy 4-member ArchesWeather ensemble (5.3% gain) while being more than six times more computationally efficient (<3.5 A100-days).

Furthermore, we demonstrate that our architecture scales predictably with increased compute. By reducing the patch size from 2 to 1, we train **Otter-XL** for approximately 10 A100-days, achieving a **14.9% gain** (fig. 1a) over IFS HRES at a 24h lead time. This result places it on par with the industrial-scale GraphCast (14.6% gain), effectively matching SOTA performance despite requiring over two orders of magnitude less compute. We present in fig. 10 the evolution of the RMSE with lead time, showing that Otter-XL maintains competitive performance even up to a 10-day lead time horizon across the majority of variables.

4.2 Probabilistic Otter Weather

While our deterministic model is highly efficient, the weather forecasting community has increasingly shifted towards probabilistic paradigms to capture predictive uncertainty. We demonstrate that Otter's architectural efficiency transfers to this setting, allowing us to also push the probabilistic skill-compute Pareto frontier. To achieve this, we evaluate two distinct pipelines: training a probabilistic model from scratch using the CRPS objective (**Otter CRPS (scratch)**) and fine-tuning an existing deterministic backbone via CRPS (**Otter CRPS**) [Diaconu et al., 2026].

Deep Ensembles through RFT. To construct our deep ensembles, we leverage the RFT phase (figs. 7 and 8). This approach differs from the concurrent work U-Cast [Cachay et al., 2026], which achieves deep ensembling by training multiple one-step CRPS fine-tuned models from different random seeds. By pushing the branching point to the RFT stage, we can efficiently generate deep ensembles using either of our two methodologies (deterministic pre-training $\rightarrow$ CRPS fine-tuning $\rightarrow$ multiple RFTs, or CRPS from-scratch $\rightarrow$ multiple RFTs). This is a critical advantage: it makes from-scratch deep ensembling computationally viable, avoiding the $>$ 20 A100-day cost per model seen in other architectures, and ensures practitioners retain full, independent control over the entire pre-training pipeline.

Furthermore, in alignment with our efficiency philosophy, we generate these ensemble members effectively "for free." Rather than training with different random seeds, we construct the ensemble using the checkpoints generated during standard RFT hyperparameter tuning (specifically, by varying the learning rates).[2] This generates a probabilistic deep ensemble as a direct byproduct of the standard development workflow, bypassing the need for full-scale retraining. To rigorously evaluate this efficiency trade-off, we extend our analysis to include ensembles generated via distinct random seeds. This approach yields a modest enhancement in predictive skill without inflating the compute budget. While both variations confirm that highly effective ensembles can be constructed under tight resource constraints, systematically identifying the optimal branching strategy across the training pipeline to maximise ensemble diversity and calibration remains an important direction for future work.

CRPS Objective. We optimise the probabilistic version of Otter Weather with the Continuous Ranked Probability Score (CRPS) Gneiting and Raftery [2007], a proper scoring rule applied to the marginal distributions of the output. We generate an ensemble of M forecasts $(X_1, \cdots X_M)$ for a given ground-truth observation y and compute the unbiased empirical CRPS estimator ('fair' CRPS following Alet et al. [2025]:

$$\text{CRPS}_{\text{fair}} = \frac{1}{M}\sum_{i=1}^{M}|X_i - y| - \frac{1}{2M(M-1)}\sum_{i=1}^{M}\sum_{j=1}^{M}|X_i - X_j|, \tag{1}$$

where the first term encourages each ensemble member to be accurate (and minimise the absolute error), while the second term encourages dispersion in the ensemble members.

Injecting Stochasticity: AdaLN vs. MC Dropout. Some source of stochasticity needs to be used to generate the M ensemble members X_i. The prevailing approach in weather and diffusion

[2]We note that we depart from the standard definition of deep ensembles from Lakshminarayanan et al. [2017], and employ a method that shares conceptual similarities to Garipov et al. [2018]. We refer to appendix B.3.5 for a more detailed comparison.

models is to inject latent noise via Adaptive Layer Normalisation (AdaLN) [Lang et al., 2026, Alet et al., 2025, Diaconu et al., 2026]. However, drawing inspiration from concurrent findings in U-Net architectures [Cachay et al., 2026], we ablate this standard AdaLN approach against a simpler, parameter-free alternative: Monte Carlo (MC) Dropout. In our Transformer-based setting, we find that MC Dropout naturally aligns with the existing dropout layers used during deterministic pre-training. Our ablations reveal that MC Dropout not only avoids the parameter overhead of AdaLN blocks but actually achieves superior rollout CRPS metrics, though it results in slightly more underdispersed ensembles, as indicated in fig. 4.

Regarding the dropout ablations, the Muon optimiser proved essential for convergence, consistently outperforming AdamW in rollout CRPS (see fig. 12), which aligns with findings from U-Cast. However, Muon-optimised ensembles are slightly more underdispersed. Ablations of the AdaLN injection strategy (fig. 13) reveal that the backbone learning rate is the primary driver of calibration (SSR), while the noise branch learning rate significantly influences the skill-spread trade-off. Generally, lower learning rates favour calibration over raw CRPS skill. Furthermore, increasing the noise embedding dimension D beyond 64 yielded negligible improvements while increasing parameter count and training latency, and higher weight decay (0.2) on the noise branch proved strictly inferior to 0.1 for both skill and calibration.

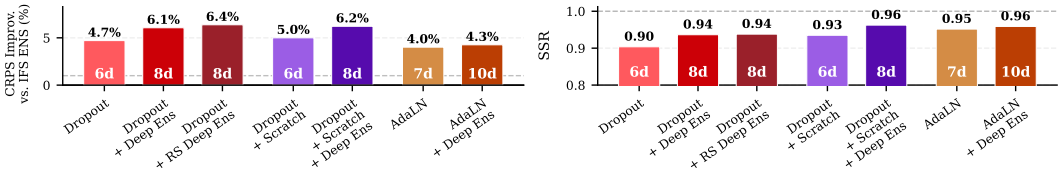
From-scratch versus Fine-tuning. We additionally compare Otter CRPS (fine-tuned from the deterministic checkpoint) to an equivalent model trained from scratch under a matched compute budget—namely, the sum of the deterministic pretraining and CRPS fine-tuning costs. While the final 6h training loss of the model trained from scratch is higher than that of the fine-tuned model (7% higher—0.29 versus 0.27), after the RFT procedure we surprisingly find that both variants achieve similar performance, with the model trained from scratch achieving a 6.2% average CRPS improvement over 1-10 days lead times as compared to 6.1% for the fine-tuned model (fig. 4). This suggests that, with our efficient architecture, both options are viable choices for obtaining a competitive probabilistic model in less than 8 A100 days, with the latter preferred if the deterministic checkpoint is already available, or when practitioners want to maintain control over the pre-training pipeline.

Figures 14 and 15 show the evolution of the CRPS improvement over IFS ENS and of the SSR with lead time up to 10 days, showing that the two methods achieve very similar CRPS metrics, but the models trained from scratch tend to show better calibration profiles—while Otter CRPS generally tends to be underdispersed, especially at the shorter lead times, the SSR becomes closer to 1 for the model trained from scratch versus the fine-tuned one. Example predictions for both probabilistic variants are available in appendix C.6.

Deep Ensemble: learning rates vs. random seeds. The standard practice for constructing deep ensembles relies on training multiple independent models with distinct random seeds [Lakshminarayanan et al., 2017]. As illustrated in fig. 4, we compare this conventional approach against our efficiency-driven method, which constructs an ensemble effectively "for free" by repurposing model checkpoints generated during standard learning rate hyperparameter tuning. Under a strict computational budget of 8 A100-days, our hyperparameter-based method yields a highly competitive 6.1% Continuous Ranked Probability Score (CRPS) improvement over the IFS ENS baseline and maintains a Spread-Skill Ratio (SSR) of 0.94. When employing the standard random seed approach (RS Deep Ens) within the exact same compute footprint, we observe a modest performance advantage, pushing the CRPS improvement to 6.4% while maintaining the 0.94 SSR. These results indicate that while explicitly varying random seeds offers slightly superior predictive skill, repurposing hyperparameter checkpoints successfully produces a calibrated, highly performant ensemble in the low-compute regime without requiring dedicated retraining runs.

Probabilistic Pareto Frontier. By combining MC Dropout with an RFT ensembling strategy, we achieve an effective probabilistic medium-range weather model at a fraction of the cost of current SOTA models. As demonstrated in fig. 1b, this approach fundamentally advances the probabilistic skill-compute Pareto frontier. Using a total budget of under 8 A100-days, the base Otter model surpasses flow-based baselines like ArchesWeatherGen while requiring less than half of their > 20 A100-day training footprint. When scaling this framework to larger compute regimes, the underlying architecture remains exceptionally performant. As mapped out in the skill-compute frontier of fig. 1b (and detailed across model configurations with Otter family in fig. 5), our scaled variant, Otter-XL, dominates existing baselines. In particular, Otter-XL (Deep Ens) proves remarkably capable: requiring only 29 A100-days of training compute, it achieves a 9.7% CRPS improvement over the IFS

ENS baseline while maintaining a Spread-Skill Ratio (SSR) of 0.95. This allows the model to deliver a nearly two-fold skill increase over comparable lightweight architectures, while simultaneously outperforming resource-intensive frontier models like GenCast by a clear 2.2% margin at a 10-fold reduction in training compute. In figs. 16 and 17, we include Otter XL in the lead-time evolution of CRPS improvement over IFS ENS and SSR up to 10 days, demonstrating that the gains from a larger model persist across all lead times, though they diminish as lead time increases.



(a) CRPS (b) SSR

Figure 4: **Performance of Probabilistic Training Strategies.** Average CRPS improvement over the IFS ensemble (a) and SSR (b) across 1–10 day lead times for three Otter CRPS variants: fine-tuning with MC Dropout, from-scratch training, and fine-tuning with AdaLN injection. For each base configuration, employing a Deep Ensemble (Deep Ens) consistently enhances skill and improves the spread.

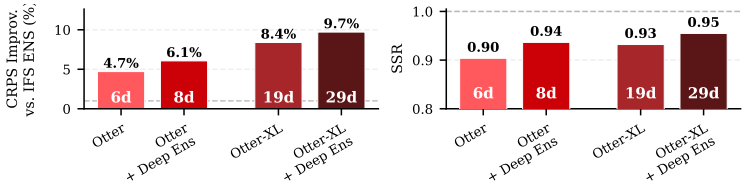


Figure 5: **Scaling behavior of probabilistic Otter models.** Side-by-side comparison of CRPS improvement against IFS ENS (%) (left) and Spread-Skill Ratio (right) across 1–10 day lead times for single-model and Deep Ensemble (Deep Ens) configurations of Otter and Otter-XL. Compute costs are annotated inside the bars in A100 days. At approximately 3.5× the computational cost, the Otter-XL models yield over a 55% relative increase in forecasting skill over their respective base Otter configurations.

5 Cross-Domain Generality

To evaluate the generalisability of our architectural insights beyond weather forecasting, we apply the Otter base architecture to the Acoustic Scattering Inclusions dataset from *The Well* benchmark [Ohana et al., 2024]. This dataset models the propagation of acoustic pressure waves through domains with varying material inclusions—a physical regime that differs fundamentally from global atmospheric dynamics. For this evaluation, we restricted our training budget to use fewer datapoints than the budget allocated per dataset by Walrus [McCabe et al., 2026], a current SOTA foundation model for this benchmark. Despite this more limited budget, Otter demonstrates strong out-of-the-box performance, achieving a superior one-step VRMSE of 0.0030 compared to the 0.0089 reported for Walrus. These gains persist across extended rollouts (see fig. 27): in the 1–20 step horizon, Otter achieves a VRMSE of 0.020 (vs. 0.043 for Walrus), and 0.113 (vs. 0.1427) in the 21–60 step horizon. Detailed training protocols and additional metrics are provided in appendix D, and some example predictions in fig. 28.

More importantly, our ablation studies (fig. 6) suggest that the core design principles identified in the weather domain can transfer effectively to this PDE setting. The Muon optimiser once again demonstrates significant efficiency gains: at a matched 10-epoch budget, reverting to standard AdamW degrades performance by 29.6%. The architectural substitutions that proved effective for weather also appear critical here, as replacing SwiGLU with GeLU (-23.1%) or removing 2D RoPE embeddings (-5.2%) cause drops in skill. Finally, in line with our earlier findings, a balanced backbone architecture consistently outperforms both shallower, wider variants and deeper alternatives. While our current investigation is limited to a single dataset, the consistency of these findings across different physical scales is encouraging. It suggests that a well-regularised, domain-agnostic

Transformer equipped with modern optimisations may provide an efficient template for modelling complex spatiotemporal dynamics. Investigating whether these design principles scale and generalise to broader PDE foundation models remains an interesting and promising direction for future work.

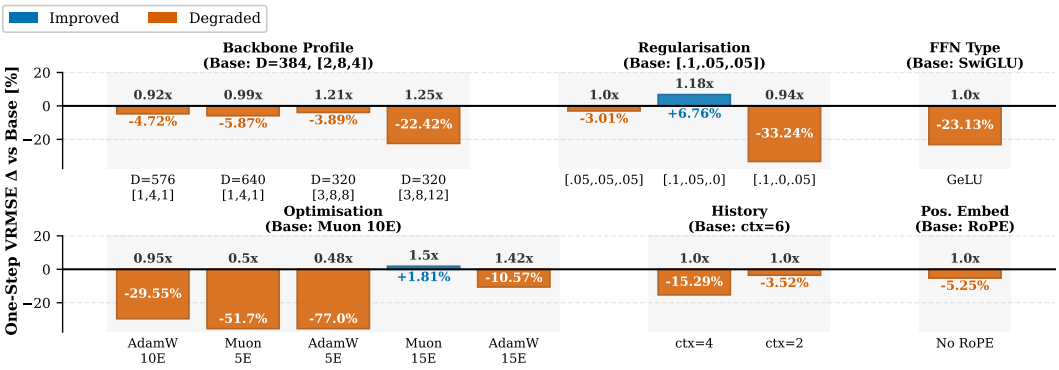


Figure 6: Ablation study on the acoustic scattering PDE task. We report the one-step VRMSE over the test set. The results confirm that findings from the weather domain (e.g., superiority of Muon, SwiGLU, RoPE, and balanced backbones) transfer to other physical systems.

6 Conclusion

We introduced Otter Weather to demonstrate that highly competitive global weather forecasting requires neither massive compute budgets nor rigid domain-specific inductive biases. By systematically integrating general-purpose advances from the LLM and vision communities, we push the skill-compute Pareto frontier for both deterministic and probabilistic modelling. Our cross-domain results on acoustic scattering PDEs further suggest that these design principles might extend broadly to spatiotemporal scientific machine learning. Importantly, we achieve these gains operating under a highly constrained training budget: under 3.5 A100-days for our deterministic model and under 8 A100-days for our probabilistic deep ensembles. This frees researchers from dependence on industry-scale compute and the architectural constraints of large foundation models, enabling task-specific training from scratch and the rapid iteration cycle needed to accelerate novel model development.

Otter Weather provides a powerful template for democratised and sovereign weather modelling, filling a critical gap in the ecosystem of data-driven forecasting. A healthy landscape must include not only expensive, general-purpose foundation models, but also lightweight, cheaply trainable alternatives. This diversity is essential for two complementary reasons. First, true scientific sovereignty requires that forecasting tools be adaptable to specific institutional needs and compute budgets without relying solely on large-scale, pre-trained models. Second, operational forecasting has long relied on ensembles of distinct models; by aggregating systems with different failure modes, we can effectively reduce correlated biases and epistemic uncertainty. Because data-driven models dramatically lower inference costs compared to traditional numerical methods, deploying these large, varied ensembles is now increasingly practical, further amplifying the need for the architectural and training diversity that accessible models like Otter provide.

Limitations and Future Work. While Otter achieves competitive performance, our approach is not without limitations (detailed in appendix E). Our findings are empirical, and we currently operate at a 1.5° resolution; while this is highly efficient, the precise trade-offs of scaling this architecture to higher resolutions warrant future exploration. Additionally, while our deep ensembles yield significant skill improvements, the optimal method for constructing them remains an exciting open question. Future work should systematically evaluate the trade-offs between training completely independent models from scratch versus branching ensemble members from a shared pre-trained checkpoint. Despite these limitations, Otter Weather provides a highly accessible, robust foundation for future research, aimed at democratising weather forecasting.

Reproducibility Statement

Detailed experimental settings are provided in the Appendix. To facilitate reproducibility, the complete implementation is available at https://github.com/cambridge-mlg/otter.

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References

Ferran Alet, Ilan Price, Andrew El-Kadi, Dominic Masters, Stratis Markou, Tom R. Andersson, Jacklynn Stott, Remi Lam, Matthew Willson, Alvaro Sanchez-Gonzalez, and Peter Battaglia. Skillful joint probabilistic weather forecasting from marginals, 2025. URL https://arxiv.org/abs/2506.10772.

Anna Allen, Stratis Markou, Will Tebbutt, James Requeima, Wessel P. Bruinsma, Tom R. Andersson, Michael Herzog, Nicholas D. Lane, Matthew Chantry, J. Scott Hosking, and Richard E. Turner. End-to-end data-driven weather prediction. *Nature*, 641(8065):1172–1179, 2025. doi: 10.1038/s41586-025-08897-0.

Zied Ben Bouallègue and the AIFS team. A new ML model in the ECMWF web charts. ECMWF AIFS Blog, 2023. URL https://www.ecmwf.int/en/about/media-centre/aifs-blog/2023/new-ml-model-ecmwf-web-charts.

Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619(7970):533–538, 2023. doi: 10.1038/s41586-023-06185-3.

Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, Jayesh K. Gupta, Kit Thambiratnam, Alexander T. Archibald, Chun-Chieh Wu, Elizabeth Heider, Max Welling, Richard E. Turner, and Paris Perdikaris. A foundation model for the Earth system. *Nature*, 641(8065):1180–1187, 2025. doi: 10.1038/s41586-025-09005-y.

Boris Bonev, Thorsten Kurth, Ankur Mahesh, Mauro Bisson, Jean Kossaifi, Karthik Kashinath, Anima Anandkumar, William D. Collins, Michael S. Pritchard, and Alexander Keller. FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale, 2025. URL https://arxiv.org/abs/2507.12144.

Salva Rühling Cachay, Miika Aittala, Karsten Kreis, Noah D Brenowitz, Arash Vahdat, Morteza Mardani, and Rose Yu. Elucidated rolling diffusion models for probabilistic forecasting of complex dynamics. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=NnRQOPzL9P.

Salva Rühling Cachay, Duncan Watson-Parris, and Rose Yu. U-Cast: A surprisingly simple and efficient frontier probabilistic AI weather forecaster. In *Forty-third International Conference on Machine Learning*, 2026. URL https://openreview.net/forum?id=XF0wkyEbuM.

Lei Chen, Xiaohui Zhong, Feng Zhang, Yuan Cheng, Yinghui Xu, Yuan Qi, and Hao Li. FuXi: a cascade machine learning forecasting system for 15-day global weather forecast. *npj Climate and Atmospheric Science*, 6:190, 2023. doi: 10.1038/s41612-023-00512-1.

Guillaume Couairon, Renu Singh, Anastase Charantonis, Christian Lessig, and Claire Monteleoni. ArchesWeather & ArchesWeatherGen: a deterministic and generative model for efficient ML weather forecasting, 2024. URL https://arxiv.org/abs/2412.12971.

Cristiana Diaconu, Miles Cranmer, Richard E. Turner, Tanya Marwah, and Payel Mukhopadhyay. Probabilistic retrofitting of learned simulators. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 24654–24701. PMLR, 2026. URL https://proceedings.mlr.press/v306/diaconu26a.html.

Dan Fu, Hermann Kumbong, Eric Nguyen, and Christopher Ré. FlashFFTConv: Efficient convolutions for long sequences with tensor cores. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 9455–9483, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/281190d87732639e63bc19dedd7d711b-Paper-Conference.pdf.

Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry Vetrov, and Andrew G Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing*

Systems, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/be3087e74e9100d4bc4c6268cdbe8456-Paper.pdf.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6185–6194, 2023.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pages 15979–15988. IEEE, 2022.

Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, Adrian Simmons, Cornel Soci, Saleh Abdalla, Xavier Abellan, Gianpaolo Balsamo, Peter Bechtold, Gionata Biavati, Jean Bidlot, Massimo Bonavita, Giovanna De Chiara, Per Dahlgren, Dick Dee, Michail Diamantakis, Rossana Dragani, Johannes Flemming, Richard Forbes, Manuel Fuentes, Alan Geer, Leo Haimberger, Sean Healy, Robin J. Hogan, Elías Hólm, Marta Janisková, Sarah Keeley, Patrick Laloyaux, Philippe Lopez, Cristina Lupu, Gabor Radnoti, Patricia de Rosnay, Iryna Rozum, Freja Vamborg, Sebastien Villaume, and Jean-Noël Thépaut. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.

Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks. https://kellerjordan.github.io/posts/muon/, 2024. Accessed: 2026-02-03.

Ryan Keisler. Forecasting global weather with graph neural networks, 2022. URL https://arxiv.org/abs/2202.07575.

Dmitrii Kochkov, Janni Yuval, Ian Langmore, Peter Norgaard, Jamie Smith, Griffin Mooers, Milan Klöwer, James Lottes, Stephan Rasp, Peter Düben, Sam Hatfield, Peter Battaglia, Alvaro Sanchez-Gonzalez, Matthew Willson, Michael P. Brenner, and Stephan Hoyer. Neural general circulation models for weather and climate. *Nature*, 632(8027):1060–1066, 2024. doi: 10.1038/s41586-024-07744-y.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Merose, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023. doi: 10.1126/science.adi2336.

Simon Lang, Mihai Alexe, Matthew Chantry, Jesper Dramsch, Florian Pinault, Baudouin Raoult, Mariana C. A. Clare, Christian Lessig, Michael Maier-Gerber, Linus Magnusson, Zied Ben Bouallègue, Ana Prieto Nemesio, Peter D. Dueben, Andrew Brown, Florian Pappenberger, and Florence Rabier. AIFS – ECMWF's data-driven forecasting system, 2024. URL https://arxiv.org/abs/2406.01465.

Simon Lang, Mihai Alexe, Mariana C. A. Clare, Christopher Roberts, Rilwan Adewoyin, Zied Ben Bouallègue, Matthew Chantry, Jesper Dramsch, Peter D. Dueben, Sara Hahner, Pedro Maciel, Ana Prieto-Nemesio, Cathal O'Brien, Florian Pinault, Jan Polster, Baudouin Raoult, Steffen Tietsche, and Martin Leutbecher. AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *npj Artificial Intelligence*, 2026. doi: 10.1038/s44387-026-00073-7.

Fanny Lehmann, Firat Ozdemir, Benedikt Soja, Torsten Hoefler, Siddhartha Mishra, and Sebastian Schemm. Finetuning a weather foundation model with lightweight decoders for unseen physical processes, 2025. URL https://arxiv.org/abs/2506.19088.

Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is scalable for LLM training, 2025. URL https://arxiv.org/abs/2502.16982.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9992–10002, 2021. doi: 10.1109/ICCV48922.2021.00986.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Ankur Mahesh, William D. Collins, Boris Bonev, Noah Brenowitz, Yair Cohen, Joshua Elms, Peter Harrington, Karthik Kashinath, Thorsten Kurth, Joshua North, Travis O'Brien, Michael Pritchard, David Pruitt, Mark Risser, Shashank Subramanian, and Jared Willard. Huge ensembles – Part 1: Design of ensemble weather forecasts using spherical Fourier neural operators. *Geoscientific Model Development*, 18(17):5575–5603, 2025. doi: 10.5194/gmd-18-5575-2025.

Michael McCabe, Payel Mukhopadhyay, Tanya Marwah, Bruno Régaldo-Saint Blancard, François Rozet, Cristiana Diaconu, Lucas Thibaut Meyer, Kaze W. K. Wong, Mohammad-Hadi Sotoudeh, Alberto Bietti, Irina Espejo Morales, Rio Alexa Fear, Siavash Golkar, Tom Hehir, Keiya Hirashima, Geraud Krawezik, François Lanusse, Rudy Morel, Ruben Ohana, Liam Holden Parker, Mariel Pettee, Jeff Shen, Kyunghyun Cho, Miles Cranmer, and Shirley Ho. Walrus: A cross-domain foundation model for continuum dynamics. In *Forty-third International Conference on Machine Learning*, 2026. URL https://openreview.net/forum?id=kcyOjXoUZu.

Tung Nguyen, Johannes Brandstetter, Ashish Kapoor, Jayesh K Gupta, and Aditya Grover. ClimaX: A foundation model for weather and climate. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 25904–25938. PMLR, 23–29 Jul 2023. URL https://proceedings.mlr.press/v202/nguyen23a.html.

Tung Nguyen, Rohan Shah, Hritik Bansal, Troy Arcomano, Romit Maulik, Veerabhadra Kotamarthi, Ian Foster, Sandeep Madireddy, and Aditya Grover. Scaling transformer neural networks for skillful and reliable medium-range weather forecasting. In *Advances in Neural Information Processing Systems*, volume 37, pages 68740–68771, 2024. doi: 10.52202/079017-2196.

Ruben Ohana, Michael McCabe, Lucas Meyer, Rudy Morel, Fruzsina J. Agocs, Miguel Beneitez, Marsha Berger, Blakesley Burkhart, Keaton Burns, Stuart B. Dalziel, Drummond B. Fielding, Daniel Fortunato, Jared A. Goldberg, Keiya Hirashima, Yan-Fei Jiang, Rich R. Kerswell, Suryanarayana Maddu, Jonah Miller, Payel Mukhopadhyay, Stefan S. Nixon, Jeff Shen, Romain Watteaux, Bruno Régaldo-Saint Blancard, François Rozet, Liam H. Parker, Miles Cranmer, and Shirley Ho. The well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 37, pages 44989–45037, 2024. doi: 10.52202/079017-1430.

Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Animashree Anandkumar. FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators, 2022. URL https://arxiv.org/abs/2202.11214.

Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044):84–90, 2025. doi: 10.1038/s41586-024-08252-9.

Stephan Rasp, Stephan Hoyer, Alexander Merose, Ian Langmore, Peter Battaglia, Tyler Russell, Alvaro Sanchez-Gonzalez, Vivian Yang, Rob Carver, Shreya Agrawal, Matthew Chantry, Zied Ben Bouallègue, Peter Dueben, Carla Bromberg, Jared Sisk, Luke Barrington, Aaron Bell, and Fei Sha. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *Journal of Advances in Modeling Earth Systems*, 16(6):e2023MS004019, 2024. doi: 10.1029/2023MS004019. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2023MS004019.

Marc Rußwurm, Konstantin Klemmer, Esther Rolf, Robin Zbinden, and Devis Tuia. Geographic location encoding with spherical harmonics and sinusoidal representation networks. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=PudduufFLa.

Noam Shazeer. GLU variants improve transformer, 2020. URL https://arxiv.org/abs/2002.05202.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017. URL https://openreview.net/forum?id=B1ckMDqlg.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing*, 568:127063, 2024. doi: 10.1016/j.neucom.2023.127063.

Christopher Subich. Efficient fine-tuning of 37-level GraphCast with the Canadian global deterministic analysis. *Artificial Intelligence for the Earth Systems*, 4(3), July 2025. ISSN 2769-7525. doi: 10.1175/AIES-D-24-0101.1.

Defa Zhu, Hongzhi Huang, Zihao Huang, Yutao Zeng, Yunyao Mao, Banggu Wu, Qiyang Min, and Xun Zhou. Hyper-connections. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=9FqARW7dwB.

A Summary Table of Related Work

We provide a comparative summary of deterministic weather models in table 1. Despite operating with the lowest compute budget, Otter Weather achieves superior performance in the medium-resolution category and remains competitive with state-of-the-art models in the high-resolution regime. Otter-XL closely approaches the performance of the best high-resolution model (GraphCast) despite using two orders of magnitude less compute.

Table 1: **Comparison of deterministic global weather models. Params** refers to the total number of parameters. **Cost** refers to training compute in A100-day equivalents (adapted from Couairon et al. [2024]). **Skill** is average RMSE improvement (%) over IFS HRES, aggregated over 24h–72h (↑).

Model	Architecture	Res.	Params	Cost	Skill %
Numerical models					
IFS HRES	Numerical	0.1°	–	–	0.00
High-resolution (< 1°)					
Pangu-Weather [Bi et al., 2023]	3D Earth-Specific Transformer	0.25°	256M	1440	5.84
FuXi [Chen et al., 2023]	Cascade U-Transformer	0.25°	1.5B	**26**	14.44
NeuralGCM (0.7°) [Kochkov et al., 2024]	Hybrid (Diff. Dynamics + NN)	0.7°	31.1M	8064	14.61
GraphCast [Lam et al., 2023]	Multi-mesh GNN	0.25°	36.7M	1344	**16.32**
Medium-resolution (> 1°)					
Keisler GNN [Keisler, 2022]	Icosahedral GNN	1.0°	6.7M	5.5	-15.27
Stormer ENS (mean) [Nguyen et al., 2024]	Vision Transformer (ensemble)	1.5°	300M	128	6.11
NeuralGCM 50×ENS [Kochkov et al., 2024]	Hybrid (Diff. Dynamics + NN)	1.4°	18.3M	7680	7.79
ArchesWeather-M×4 [Couairon et al., 2024]	Swin U-Net (ensemble)	1.5°	84M	20	8.86
Otter Weather (Ours)	**Constant-Width 2D Swin U-Net**	1.5°	1.5B	**3**	**11.54**
Otter-XL (Ours)	**Constant-Width 2D Swin U-Net**	1.5°	1.5B	10	**16.12**

[2]We are reporting the parameter count for Lang et al. 2026 (the probabilistic version of the model) as we could not find the corresponding count for the deterministic model.

B Detailed Architecture and Training Details

This appendix provides a detailed specification of our deterministic model architecture.

B.1 Input Representation

Normalisation and Flattening. Conditioned on L historical states, the input tensor to the network has dimensions (B, L, H, W, C), where B denotes the batch size, T the temporal context length, $H = 121$ and $W = 240$ the spatial resolution (corresponding to a 1.5° grid), and $C = 87$ the number of modelled variables. We first apply standard normalisation using variable-specific statistics (mean and standard deviation) computed over the training corpus. Subsequently, we flatten the temporal and channel dimensions to obtain a tensor of shape $(B, H, W, L \cdot C)$. This reshaping allows the 2D backbone to process temporal history as distinct input channels, enabling the learning of spatial-temporal correlations through channel mixing mechanisms.

Temporal Embeddings. Atmospheric dynamics are characterised by phenomena occurring across a wide range of temporal scales, many of which exhibit periodic behaviour. To capture these cyclic dependencies, we follow Bodnar et al. 2025 and generate Fourier-based temporal embeddings derived from the time elapsed t (in hours) relative to a fixed reference date (January 1, 1979). Specifically, for a set of $N = 16$ scales s_i spaced logarithmically between ζ_{min} and ζ_{max}, we compute the embedding $\text{Emb}(t)$ as:

$$\text{Emb}(t) = \bigoplus_{i=1}^{N} \left[\sin\left(\frac{2\pi t}{s_i}\right), \cos\left(\frac{2\pi t}{s_i}\right) \right]. \tag{2}$$

We select $\lambda_{\text{min}} = 3.0$ and $\lambda_{\text{max}} = 8760$ (corresponding to the number of hours in a year). As argued in Bodnar et al. 2025, this multi-scale embedding allows the model to resolve critical temporal signals, such as diurnal cycles, weekly patterns, and seasonal variations.

Residual Learning. Consistent with recent advances in the field [Lam et al., 2023, Nguyen et al., 2024], we formulate the prediction task as learning the residual update (tendency) rather than the full future state. The prediction for the next step is defined as an additive update to the most recent context state X_t:

$$\hat{X}_{t+\Delta t} = X_t + \text{Backbone}(X_{t-T:t}, \text{Emb}(t + \Delta t)).$$

B.2 Backbone: Swin U-Net.

The core backbone of our model is a Swin Transformer [Liu et al., 2021] adapted into a symmetric encoder-decoder (U-Net) architecture. This hierarchical design allows the model to capture multi-scale atmospheric phenomena.

B.2.1 Patch Partitioning (Tokeniser).

The first step in the backbone is to transform the input tensor into a sequence of discrete tokens. We implement this using a Patch Partition layer, implemented as a strided convolution. Given an input tensor of shape $(B, H, W, L \cdot C)$, we project local patches into a latent embedding dimension D. By setting the stride P equal to the patch size, we partition the image into a grid of non-overlapping windows, resulting in a representation of shape $(B, H/P, W/P, D)$.

B.2.2 Positional Embeddings.

We explore multiple strategies to inject positional information into the architecture.

Absolute Fourier Spatial Embeddings. We follow the strategy employed in Bodnar et al. 2025 to encode absolute positional information through Fourier Embeddings. Given a number $N_s = 128$ of scales, the first $N_s/2$ are used for latitude (ϕ) information, and the next $N_s/2$ for longitude (λ). Each coordinate is independently projected into Fourier features as described in appendix B.1. We

use $\zeta_{\text{min}} = 0.1$ and $\zeta_{\text{max}} = 720$. These feature vectors are broadcast to form 2D spatial grids and concatenated along the channel dimension. Finally, a learnable linear layer projects the concatenated features to the model's token dimension D before adding them to the input tokens:

$$E_{pos} = \text{Linear}(\text{Emb}(\phi) \oplus \text{Emb}(\lambda))$$

Alternative: Spherical Harmonics Embedding. To contrast the use of standard positional embeddings with more specialised versions that strictly respect spherical symmetry, we also investigate the use of spherical harmonics embeddings following Rußwurm et al. 2024. We pre-compute the real spherical harmonic basis functions $Y_\ell^m(\theta, \phi)$ up to a maximum degree $L_{\text{max}} = 20$. For every spatial location (h, w), we generate a dense feature vector containing all harmonic terms for degrees $0 \leq \ell \leq L_{\text{max}}$ and orders $-\ell \leq m \leq \ell$. This results in $(L_{\text{max}} + 1)^2 = 441$ unique geometric features per pixel. These raw harmonic features are cached and passed through a learnable linear projection to match the token dimension D before being added to the input tokens.

Rotary Positional Embeddings (RoPE). While absolute embeddings provide global context, we also investigate the use of relative positional embeddings. We employ Rotary Positional Embeddings (RoPE) [Su et al., 2024] within the self-attention mechanism of each Swin block.

We use a 2D extension of RoPE where the spatial indices (i, j) of a token rotate the query and key vectors in the complex plane. This allows the attention mechanism to depend only on the relative distance $\Delta = (i_1 - i_2, j_1 - j_2)$ between tokens.

B.2.3 Hierarchical Encoder-Decoder Structure

The processed tokens pass through a symmetric U-Net structure composed of Swin Transformer Stages.

We adopt a "constant-width" U-Net configuration, maintaining the same channel capacity across all resolutions. The network consists of 2 downsampling levels (3 stages total, including the bottleneck). The hyperparameters for the base configuration are:

- Embedding Dimension (D): 1536.
- Encoder Depth: 2 Stages with block counts $[2, 8]$.
- Bottleneck Depth: 4 Blocks.
- Decoder Depth: 2 Stages with block counts $[8, 2]$ (symmetric to encoder).
- Attention Heads: 16 heads per block (dimension per head $d_k = 1536/16 = 96$).
- Window Size: 15×15.

This results in a symmetric depth profile of (2, 8, 4, 8, 2) blocks. The downsampling operations reduce the spatial resolution by a factor of 2 at each stage (via 2×2 strided convolution) while maintaining the channel dimension constant at D. The skip connections use complex fusion (concatenation followed by a 1×1 convolution) rather than simple summation.

B.2.4 Swin Transformer Block and Feed-Forward Network

The fundamental building block of our architecture is the Swin Transformer Block [Liu et al., 2021]. To balance local feature extraction with long-range dependencies, each block consists of two attention operations:

- **Window Attention (W-MSA).** The first unit partitions the input state into non-overlapping local windows of size $W \times W$ (where $W = 15$ in the base configuration) and computes self-attention independently within each window.
- **Shifted Window Attention (SW-MSA).** The second unit enables cross-window information exchange by shifting the window partitioning by $(\lfloor \frac{W}{2} \rfloor, \lfloor \frac{W}{2} \rfloor)$ before computing self-attention.

SwiGLU Feed-Forward Network. We replace the standard GELU-MLP used in the original Swin architecture with a SwiGLU variant, a modification shown to improve stability in large-scale models [Shazeer, 2020, Couairon et al., 2024]. The activation is defined as:

$$\text{SwiGLU}(x) = (\text{SiLU}(xW_1) \otimes xW_2)W_3$$

where $\otimes$ denotes element-wise multiplication. To maintain parameter parity with standard Transformer MLPs (typically expansion ratio 4), we reduce the hidden dimension expansion to $\approx \frac{8}{3}$ ($2.67\times$). Additionally, we enforce hardware-aware alignment by padding the hidden dimension to the nearest multiple of 128, ensuring optimal memory tiling for GPU Tensor Cores.

B.3 Additional Training Details

B.3.1 Loss.

The deterministic model is trained to minimise a latitude-weighted Root Mean Squared Error (RMSE) loss, a standard objective in global forecasting that accounts for projection distortion by ensuring that each pixel's loss contribution is proportional to its area. Variables are additionally weighted in proportion to their atmospheric pressure level. Following Lam et al. 2023, Couairon et al. 2024, surface-level variables are assigned weights of 1, with the exception of the 2m temperature, which is given a weight of 10. Mathematically, we can express the loss computed per variable v and at lead time t as:

$$\mathcal{L}_v^t = \sqrt{\frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} w(\phi_i), \left(\hat{X}_{i,j,v}^t - X_{i,j,v}^t\right)^2} \tag{3}$$

where $w(\phi_i) = \cos \phi_i / \frac{1}{H} \sum_{k=1}^{H} \cos \phi_k$ downweights grid cells toward the poles, and ϕ_i is the latitude of row i over a uniform $H \times W$ grid. The aggregated loss sums over all target variables and pressure levels v, averaging over lead times $t = 1, \ldots, T$:

$$\mathcal{L} = \sum_v \frac{1}{T} \sum_{t=1}^{T} \mathcal{L}_v^t \tag{4}$$

For CRPS training, we use a similar weighting, but apply it to the 'fair' CRPS (defined in eq. (1)). During training, we always evaluate the CRPS based on $M = 2$ ensemble members.

B.3.2 Optimisation.

We train Otter Weather using the Muon optimiser [Jordan et al., 2024]. By treating 2D parameters natively as matrix objects and transforming the momentum from both sides, Muon orthogonalises the updates—encouraging learning along non-dominant directions and accelerating convergence [Liu et al., 2025]. In our setting, however, this matrix-level operation makes each Muon step substantially more expensive than an AdamW step, a gap amplified by the small batch sizes typical in weather modelling. We offset this overhead by using gradient accumulation (4 mini-batches per optimisation step), which increases the effective batch size and amortises Muon's per-step cost. With this modification, we find that Muon improves training convergence. Notably, we observe that while the *presence* of learning rate decay is critical for performance, the specific decay profile (e.g., linear vs. cosine) is of less importance.

B.3.3 Hardware & Compute Budget.

We detail below the hardware used in this investigation. Our cluster includes nodes equipped with:

- NVIDIA RTX 6000 (Blackwell Server Edition), 128 vCPUs, 96GB VRAM;
- NVIDIA A100-PCIe (80GB), 24 vCPUs, 80GB VRAM;
- NVIDIA RTX 6000 (Ada Generation), 28 vCPUs, 48GB VRAM.

To ensure fair comparison and reproducibility, all computational costs reported in the main text are normalised to the equivalent of A100-80GB GPU hours. Furthermore, regardless of the node capacity,

all experiments are strictly conducted in a single-GPU regime, ensuring that our results are representative of performance achievable on accessible hardware without requiring multi-GPU distributed training strategies. Peak VRAM achieved during training of the models using the deterministic Base configuration was 46.5GB, and inference time for one autoregressive step takes approximately 0.48s.

B.3.4 Learning Rate Schedule.

Deterministic model. We employ a cosine decay learning rate schedule with a linear warmup of 500 steps.

- **Pretraining:** The learning rate increases linearly to a maximum of 2×10^{-4} during warmup, followed by a cosine decay to a minimum of 2×10^{-6}.
- **Rollout Fine-tuning (RFT):** To ensure stability, we initialise the fine-tuning schedule at the final learning rate of the pretraining phase and decay to 0 following a cosine schedule.

Probabilistic model: Otter CRPS.

- **One-step CRPS pretraining:** Starting from the pretrained deterministic base Otter model, we employ a cosine decay learning rate schedule with a linear warmup of 500 steps. The learning rate increases linearly to a maximum of 2×10^{-4} during warmup, followed by a cosine decay to a minimum of 2×10^{-6}.
- **Rollout Fine-tuning (RFT):** For this stage, we employ a cosine decay learning rate schedule without a warmup, decaying to a minimum of 10^{-9}. For the best single **Otter CRPS** model, we used an initial learning rate of 5×10^{-6}. To create the **Otter CRPS (Deep Ens)**, we combine three models fine-tuned with initial learning rates of 10^{-6}, 2×10^{-6}, and 5×10^{-6}.

Probabilistic model: Otter CRPS (scratch).

The **Otter CRPS (scratch)** model is initialized from scratch rather than the pretrained base. It utilises the exact same learning rate schedules for both the one-step CRPS pretraining and Rollout Fine-tuning stages as described above, differing only in the number of epochs for which the models are trained.

Probabilistic model: Otter CRPS with AdaLN.

- **One-step CRPS pretraining:** We apply different optimisers and learning rate schedules depending on the parameter group. For the newly introduced parameters that generate AdaLN conditioning from sampled noise, we use AdamW [Loshchilov and Hutter, 2019] with a 500-step linear warmup, peaking at 2×10^{-4} before applying cosine decay down to 2×10^{-6}. Conversely, the pretrained parameters inherited from the deterministic base model are optimised using Muon [Jordan et al., 2024]. This group follows a cosine decay schedule with a longer 3000-step linear warmup and a smaller peak learning rate of 5×10^{-6}, also decaying to 2×10^{-6}.
- **Rollout Fine-tuning (RFT):** For both parameter groups, we use the Muon [Jordan et al., 2024] optimiser. We employ a cosine decay learning rate schedule decaying from 2×10^{-6} to 10^{-9} with no warmup.

B.3.5 Training Duration & Curriculum.

Deterministic model.

We train the base model for a total of 20 epochs using ERA5 reanalysis data spanning 1979–2019. With a gradient accumulation factor of 4, this results in approximately 75,000 optimisation steps.

For RFT, we restrict the dataset to the period from 2007 onwards and train for a single epoch (approx. 19,000 steps). We employ a curriculum learning strategy to stabilise training: the forecast horizon begins at 24 hours (4 autoregressive steps) and is gradually extended to 72 hours (12 steps). This extension occurs in 8 equal intervals distributed across the training duration.

Probabilistic model: Otter CRPS.

Figure 7 illustrates the training pipeline used to obtain the **Otter CRPS** and **Otter CRPS (Deep Ens)** models. It highlights the computational cost (in A100 days) and the relative performance improvements (ΔCRPS) over IFS ENS achieved after each subsequent fine-tuning stage.

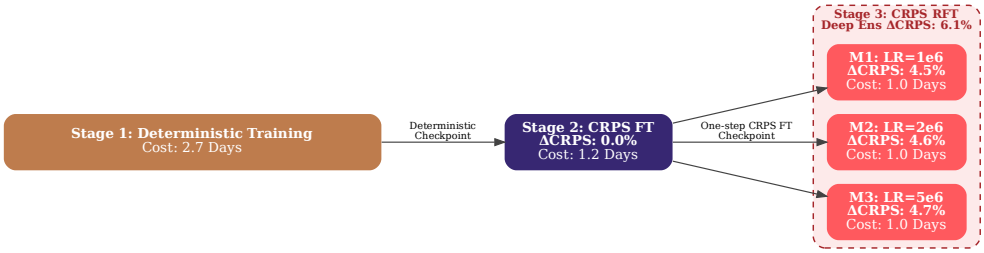


Figure 7: Overview of the training pipeline for Otter CRPS models, detailing the computational cost per stage and the progressive CRPS improvements (Δ CRPS) over IFS ENS across a 1–10 day lead time. The width of each bar is proportional to the computational time required for that stage.

Starting from the initial deterministic training phase (**Stage 1**, requiring 2.7 days), we apply a one-step CRPS fine-tuning (**Stage 2**, 1.2 days). For this stage, we fine-tune the model with the CRPS objective for 5 epochs using the same 1979–2019 ERA5 dataset used for the base model. We use an effective batch size of 16 with a gradient accumulation factor of 8.

In the final Rollout Fine-Tuning (RFT) phase (**Stage 3**), we restrict the dataset to the 2007–2019 period and apply the exact same curriculum learning strategy utilised for the deterministic model. To form a deep ensemble, the pipeline branches to train three independent models for the equivalent of 1-A100 days each, using varied learning rates (10^{-6}, 2×10^{-6}, and 5×10^{-6}). While these individual RFT models already deliver strong performance gains over IFS ENS (between 4.5% and 4.7%), aggregating their predictions as the final **Otter CRPS (Deep Ens)** provides a further substantial boost, reaching a 6.1% overall improvement in ΔCRPS.

We note that our methodology of achieving deep ensembles departs from classical deep ensembling [Lakshminarayanan et al., 2017], which typically generates members using identical hyperparameters and distinct random seeds. In contrast, we induce ensemble diversity by rollout fine-tuning each member with different hyperparameters. In this regard, our method shares closer conceptual ties with the fast ensembling approach of Garipov et al. [2018]. They construct an ensemble by employing a cyclical learning rate schedule and saving intermediate checkpoints throughout training, thereby exploring the local parameter space. We do not strictly adhere to this checkpointing procedure. Because our RFT relies on a progressive curriculum—gradually increasing the temporal unroll up to a full 3-day rollout horizon—we let ensemble member complete an entire RFT round. Consequently, rather than saving checkpoints throughout a single optimisation run, we generate the ensemble through distinct fine-tuning rounds with varied learning rates, and check that these final models do indeed achieve similar performance (i.e., between 4.5-4.7% improvement over IFS ENS). Investigating ensembling strategies that more closely adhere to the cyclical checkpointing procedure of Garipov et al. [2018] remains a promising avenue for future work.

Probabilistic model: Otter CRPS (scratch).

Figure 8 details the training pipeline for **Otter CRPS (scratch)**, highlighting both the computational cost (in A100-days) and the cumulative performance gains (ΔCRPS) over the IFS ENS baseline at each stage. In the initial phase, the model is trained from scratch with a CRPS loss for 15 epochs on the 1979–2019 ERA5 dataset, matching the deterministic model's data configuration. This stage employs an effective batch size of 16 and a gradient accumulation factor of 8. Once the base CRPS checkpoint is obtained, the pipeline advances to the Rollout Fine-Tuning (RFT) phase, which is executed identically to the **Otter CRPS** model.

Probabilistic model: Otter CRPS with AdaLN.



Figure 8: Overview of the training pipeline for Otter CRPS (scratch) models, detailing the computational cost per stage and the progressive CRPS improvements (Δ CRPS) over IFS ENS across a 1–10 day lead time. The width of each bar is proportional to the computational time required for that stage.

The training pipeline for the **Otter** model equipped with AdaLN closely mirrors the process outlined in fig. 7. Stage 2 (CRPS fine-tuning) is executed for 5 epochs on the 1979–2019 ERA5 dataset. To obtain a single base model, we perform Rollout Fine-Tuning (RFT) for a single epoch on the 2007–2019 ERA5 data, employing the same curriculum learning strategy used for the deterministic model. To construct a deep ensemble version of the AdaLN model, we then independently train three separate models targeting different forecasting horizons: 24 hours (4 autoregressive steps), 48 hours, and 72 hours. However, because this AdaLN variant ultimately underperformed the model utilising MC Dropout (see fig. 4), we did not pursue further deep ensemble configurations akin to those used for **Otter CRPS** (i.e., utilising the same RFT horizon and varying the learning rates).

C Additional Results and Evaluation Details

C.1 Additional Evaluation Details

As mentioned in the main text, we follow the methodology in Couairon et al. 2024 and provide an aggregated skill score relative to a base model (IFS HRES for deterministic models, IFS ENS for probabilistic). This is computed as follows:

$$\text{metric-ss}(\text{model}) = \frac{1}{|\mathcal{V}|} \sum_{v} \left(1 - \frac{\text{metric}_v(\text{model})}{\text{metric}_v(\text{ref})}\right), \tag{5}$$

where $\mathcal{V}$ represents the set of key headline variables we use (Z500, Q700, T850, U850, V850, T2m, SP, U10m, and V10m), $\text{metric}_v(\text{model})$ is the metric achieved by a specific model on the variable v, and ref refers to the reference model. In the deterministic case, the metric is RMSE and the reference model is IFS HRES; in the probabilisic case, the metric is the 'fair' CRPS and the reference model is IFS ENS. For a fair comparison to some of the other AI models, we only consider the common lead times (i.e., multiples of 24h as ArchesWeatherGen uses 24h as the lead time).

C.2 Base Configuration

The **Base** configuration serves as the primary reference point for all ablation studies presented in section 4.1. This architecture was established through preliminary experimentation to provide a strong balance between predictive skill and computational efficiency. The specific hyperparameters are detailed below:

- **Embedding Dimension:** $D = 1536$;
- **Depth Profile (U-Net):** $[2, 8, 4]$ blocks per stage;
- **Attention Heads:** 16;
- **FFN Activation:** SwiGLU;
- **Positional Embeddings:** Absolute (Fourier) + Rotary (RoPE);

- **Regularisation:** Weight decay 0.15, Dropout 0.1, DropPath 0.1;
- **Optimiser:** Muon (momentum $\mu = 0.95$);
- **Learning Rate:** Cosine decay (max 2×10^{-4}) with a linear warmup of 500 steps;
- **Training Budget:** 20 epochs with a gradient accumulation factor of 4 (approx. $75,000$ optimisation steps).

C.3 Additional Deterministic Results

C.3.1 Additional short-horizon metrics

While fig. 3 presents ablation results averaged over the 6h to 24h window, we also report the metrics specifically at a 24-hour lead time for the various configurations. The overall trends remain consistent across both evaluation settings.

Table 2: Ablation study of architectural, regularisation, and training components. **Compute Cost** is relative to the baseline configuration ($1.0\times$). Performance is reported as relative RMSE improvement over the base configuration (↑) at 24h lead time and does not involve RFT.

CONFIGURATION / VARIATION	COMPUTE COST VS BASE ↓	24H RMSE IMPROVEMENT ↑
Backbone Profile *(Dim D, Swin Blocks Per Stage):*		
$D = 1536, [2, 8, 4]$ (BASE)	–	–
$D = 1280, [3, 12, 8]$	$1.13\times$	0.17%
$D = 2304, [1, 4, 1]$	$1.00\times$	-0.05%
Regularisation *[Weight Decay (WD), Dropout, DropPath]:*		
$[0.15, 0.10, 0.10]$ (BASE)	–	–
$[0.15, 0.00, 0.10]$ (NO DROPOUT)	$1.00\times$	-17.61%
$[0.20, 0.10, 0.10]$ (HIGHER WD)	$1.00\times$	0.00%
$[0.10, 0.10, 0.10]$ (LOWER WD)	$1.00\times$	0.14%
$[0.15, 0.10, 0.0]$ (NO DROPPATH)	$1.09\times$	-0.96%
$[0.15, 0.20, 0.20]$ (HIGH REG)	$0.93\times$	-0.51%
Window Size*:*		
$W = 15$ (BASE)	–	–
$W = 5$	$1.02\times$	-1.03%
Patch Size*:*		
$PS = 2$ (BASE)	–	–
$PS = 1$ (OTTER-XL)	$3.25\times$	5.28%
FFN Type*:*		
SWIGLU (BASE)	–	–
GELU	$1.00\times$	-2.28%
Attention Masking*:*		
NO ATTENTION MASKING (BASE)	–	–
WITH MASKING	$1.07\times$	0.17%
Positional Embeddings*:*		
SPHERICAL + ROPE (BASE)	–	–
ABSOLUTE + ROPE	$1.00\times$	0.18%
SPHERICAL ONLY	$0.99\times$	-1.18%
Optimisation & Dynamics*:*		
MUON (20 EPOCHS) (BASE)	–	–
MUON (30 EPOCHS)	$1.50\times$	0.23%
ADAMW (35 EPOCHS)	$1.58\times$	0.14%
ADAMW (23 EPOCHS)	$1.00\times$	-0.24%
MUON (15 EPOCHS)	$0.75\times$	-0.02%
ADAM (18 EPOCHS)	$0.8\times$	-0.75%
History*:*		
4 STATES (BASE)	–	–
3 STATES	$1.00\times$	-0.20%
2 STATES	$1.00\times$	-0.65%

C.3.2 Longer-horizon evaluations

We further analyse the temporal evolution of RMSE skill relative to IFS HRES in fig. 9. We compare base Otter and Otter-XL against two representative deterministic baselines: GraphCast and ArchesWeather-4M. While the Otter models and GraphCast share a similar fine-tuning schedule with a rollout horizon of 3 days, ArchesWeather-4M uses a more extensive protocol extending to 4 days. In terms of computational efficiency, base Otter Weather is drastically more accessible: GraphCast requires a training budget approximately $450\times$ larger, while Arches-4M demands over $6\times$ the resources. Even our higher-performance configuration, Otter-XL, remains highly efficient—operating at just below $\approx 1\%$ of GraphCast's compute budget and requiring only 63% of the cost of Arches-4M.

We observe that all ML-based methods generally outperform the NWP baseline across the majority of variables and lead times. A notable exception is 2m temperature (T2M) at short lead times, where the higher resolution of the NWP model likely provides an advantage. Compared to GraphCast (navy), Otter Weather (coral) exhibits lower skill at short lead times, though this gap diminishes as the rollout horizon extends. Impressively, Otter-XL (dark red) significantly closes this margin: it remains highly competitive at short lead times and outperforms GraphCast at longer horizons. Crucially, this is achieved with a computational budget two orders of magnitude smaller, underscoring the efficiency of our approach.

Against ArchesWeather-4M (dark green), base Otter (coral) generally shows superior performance at short lead times, particularly on wind-related variables. Otter-XL overperforms by a wide margin up to ≈ 7 day lead time. While ArchesWeather-4M shows improved retention of skill at longer horizons—likely a direct consequence of its extended 4-day rollout fine-tuning—it is widely recognised that deterministic methods suffer from over-smoothing at these ranges. Rollout fine-tuning can only partially mitigate this intrinsic limitation; consequently, we view the exploration of probabilistic methods as the most effective strategy for long-horizon forecasting and a primary direction for future work.

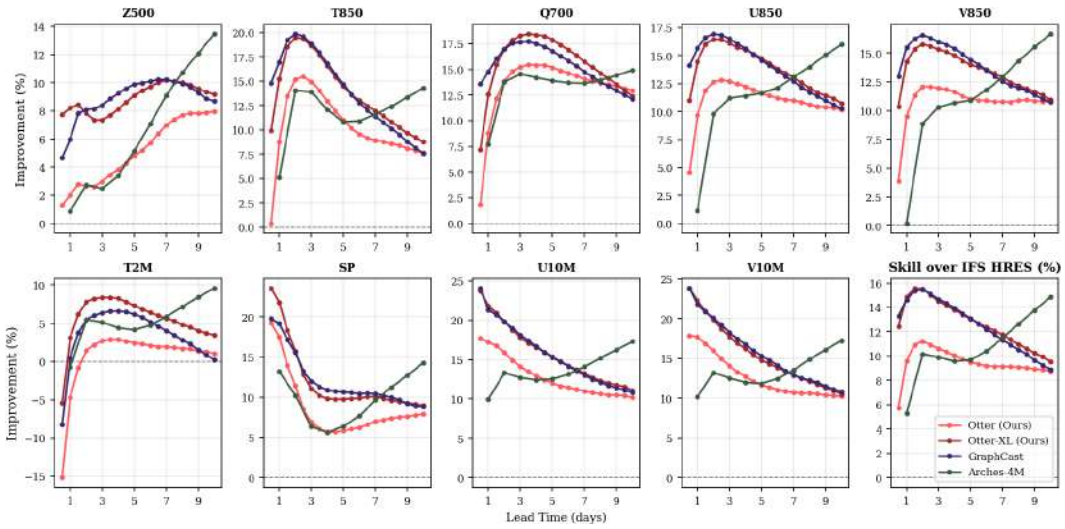


Figure 9: RMSE skill over IFS HRES comparison for lead times up to 10 days for Otter, Otter-XL, GraphCast, and ArchesWeather-4M. The last plot shows the percentage improvement over IFS HRES averaged over the headline variables.

C.4 Additional Probabilistic Results

As mentioned in appendix C.1, for the probabilistic evaluation we use the 'fair' CRPS as the evaluation metric. When comparing to other AI models, we report performance on a 1-10 day lead time horizon, computed based on 5 ensemble members. For model selection within the Otter CRPS class, we evaluate average performance between 24–72h lead times based on 10 ensemble members. Finally, our Deep Ensemble results are derived from three distinct models, each generating three ensemble members (yielding 9 members in total). Because the fair CRPS provides an unbiased estimate of the

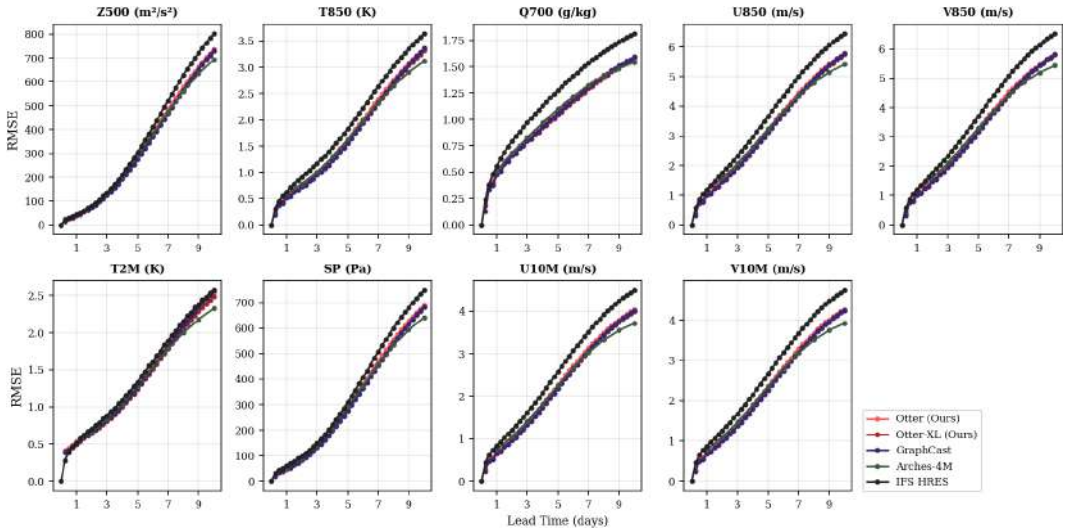


Figure 10: RMSE comparison for lead times up to 10 days. Lines represent IFS HRES (NWP baseline, black), Otter Weather (ours, coral), our more expensive Otter-XL variant (dark red), and GraphCast (navy). Otter Weather consistently outperforms the NWP baseline across most variables and lead times, remaining competitive with state-of-the-art models despite using orders of magnitude less compute. Otter-XL is competitive with GraphCast.

true CRPS, the reported metrics are largely invariant to ensemble size. This is empirically supported in fig. 11, which demonstrates that the CRPS remains stable regardless of the number of members used for evaluation.

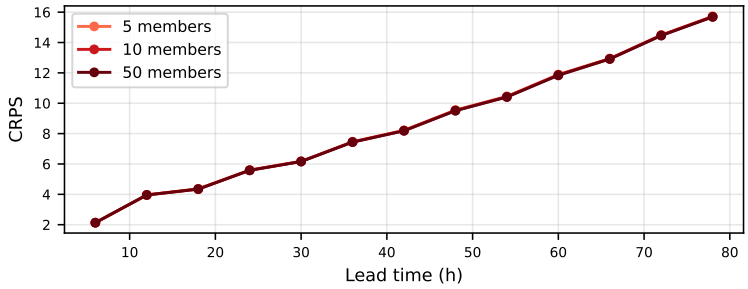


Figure 11: Evolution of the 'fair' CRPS with lead time for different number of ensemble members: [5, 10, 50]. The number of ensemble members does not affect the CRPS significantly given it is an unbiased estimator.

C.4.1 Additional results for Dropout

For the MC Dropout strategy, we study three learning rate settings (5e-6, 2-5, 2-4), and two different optimisers (AdamW and Muon). fig. 12 illustrates the CRPS skill improvement for 1–3 day lead times (left) alongside the Spread-Skill Ratio (right). The Muon optimiser consistently achieves superior CRPS metrics compared to AdamW; however, this performance gain comes at the expense of calibration, with Muon-trained models exhibiting higher levels of underdispersion in their ensembles.

C.4.2 Additional results for AdaLN

We evaluate the sensitivity of the AdaLN noise injection mechanism by varying the noise dimension D, learning rates, and weight decay. Aside from the noise branch, we use zero weight decay for the backbone and omit dropout and stochastic depth. We use the Muon optimiser for the backbone and the AdamW for the noise branch. Key findings include:

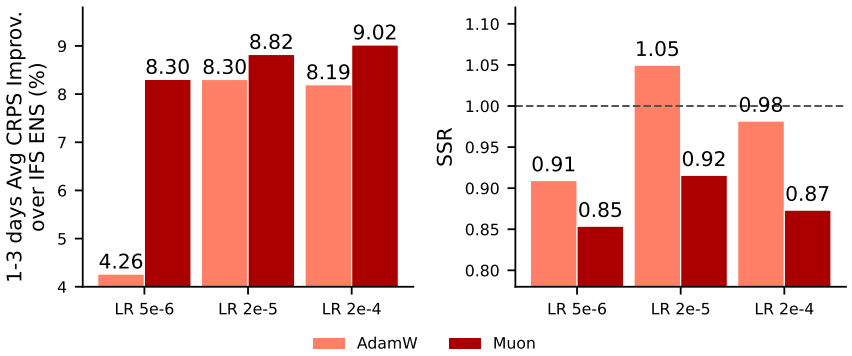


Figure 12: Skill improvement over 1–3 day lead times (left) and Spread-Skill Ratio (right) for three learning rates and two optimisers. While the Muon optimiser consistently yields superior CRPS metrics, it leads to increased underdispersion compared to AdamW-based variants.

- The noise branch learning rate dictates calibration: We observe a monotonic relationship where lower learning rate for the noise branch yields better-calibrated ensembles (SSR closer to 1.0). However, CRPS improvement is non-monotonic; 5×10^{-6} leads to the best performance, with 10^{-5} being too aggressive and degrading both skill and spread.
- Skill-calibration trade-off: The best model in terms of CRPS performance is underdispersed, whereas the better calibrated model show a 0.45% reduction in CRPS improvement.
- Backbone learning rate tuning improves SSR efficiently: Increasing the backbone learning rate from 2×10^{-4} to 5×10^{-4} improves the SSR from 0.932 to 0.957 with a negligible CRPS penalty.
- Marginal impacts of weight decay and D: Higher weight decay (0.2) is strictly inferior to 0.1. Furthermore, reducing D from 128 to 64 results in a marginal CRPS loss, albeit coming at an increased computational cost and parameter count.

The model we use for the AdaLN injection mechanism in the main text is $D = 128, 5{\times}10^{-6}/2{\times}10^{-4}$, weight decay of 0.1.

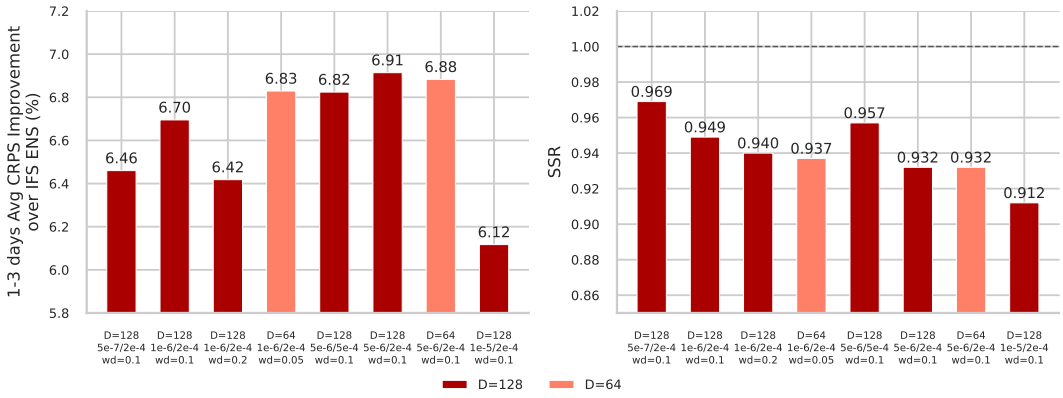


Figure 13: Skill improvement over 1–3 day lead times (left) and Spread-Skill Ratio (right) for the AdaLN noise injection strategy. The labels indicate the noise embedding dimension D, the backbone/noise branch learning rates, and the weight decay used in the noise branch.

C.4.3 From-scratch versus Fine-tuning

To further evaluate the probabilistic characteristics of our proposed models, we analyse the CRPS improvement over the IFS Ensemble alongside the Spread-Skill Ratio (SSR) across lead times up to 10 days in figs. 14 and 15. These lead to the following main observations:

- **Benefits of Deep Ensembling:** Deep ensembling yields a consistent performance boost for the Otter CRPS models across all evaluated atmospheric variables. These gains are achieved at a minimal computational cost—in our case, the ensemble members were generated by varying the learning rate during a standard hyperparameter sweep for the RFT procedure (requiring only 1 A100 day per run). Because practitioners routinely conduct such hyperparameter explorations when adapting models, they naturally generate multiple distinct checkpoints. Consequently, deep ensembling can be leveraged as a highly practical byproduct of the standard development workflow, utilising these already-available models to improve probabilistic skill without incurring dedicated ensemble training overhead.
- **Calibration and Underdispersion:** While the differences in overall CRPS improvement between the fine-tuned Otter CRPS model and Otter CRPS (scratch) are relatively small, evaluating the SSR highlights distinct calibration behaviour. The fine-tuned Otter CRPS model exhibits a tendency toward underdispersion (falling below the ideal 1.0 ratio) for certain variables. In contrast, Otter CRPS (scratch) and in particular its deep ensemble variant mitigate the underdispersion to some extent, maintaining an SSR closer to 1 for the majority of variables.
- **Comparisons to Baselines:** When benchmarking against SOTA baselines, a performance gap remains between the Otter variants and GenCast. However, Otter CRPS outperforms ArchesWeatherGen, particularly in the short-to-medium range up to approximately 6 days of lead time. We hypothesise that this specific performance profile—excelling at earlier lead times—may be a consequence of the specific RFT schedule utilised during our training process.

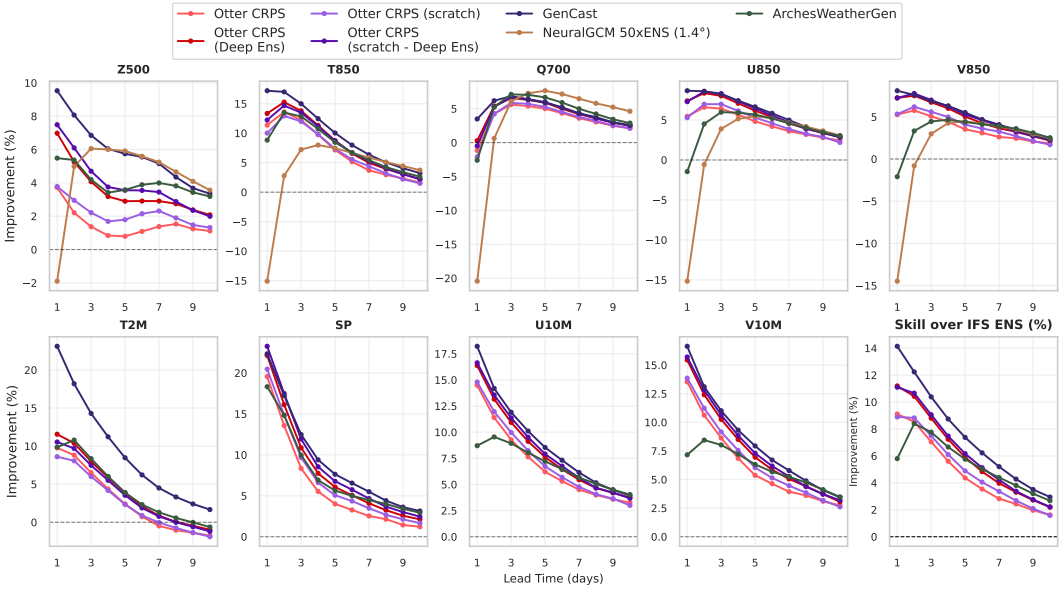


Figure 14: CRPS skill over IFS ENS comparison for lead times up to 10 days for Otter CRPS, Otter CRPS (Deep Ens), Otter CPRS (scratch), Otter CRPS (scratch - DeepEns), GenCast, and ArchesWeatherGen, and NeuralGCM. The last plot shows the percentage improvement over IFS ENS averaged over the headline variables.

C.5 Scaling the Probabilistic Model

In figs. 16 and 17 we show the CRPS improvement over IFS ENS and SSR as a function of lead time for up to 10 days, including the Otter CRPS XL variant (patch size 1). These figures confirm that the improvements from the more compute-intensive variant are not confined to short lead times but are maintained throughout the full 10-day period, with gains diminishing gradually as lead time increases.

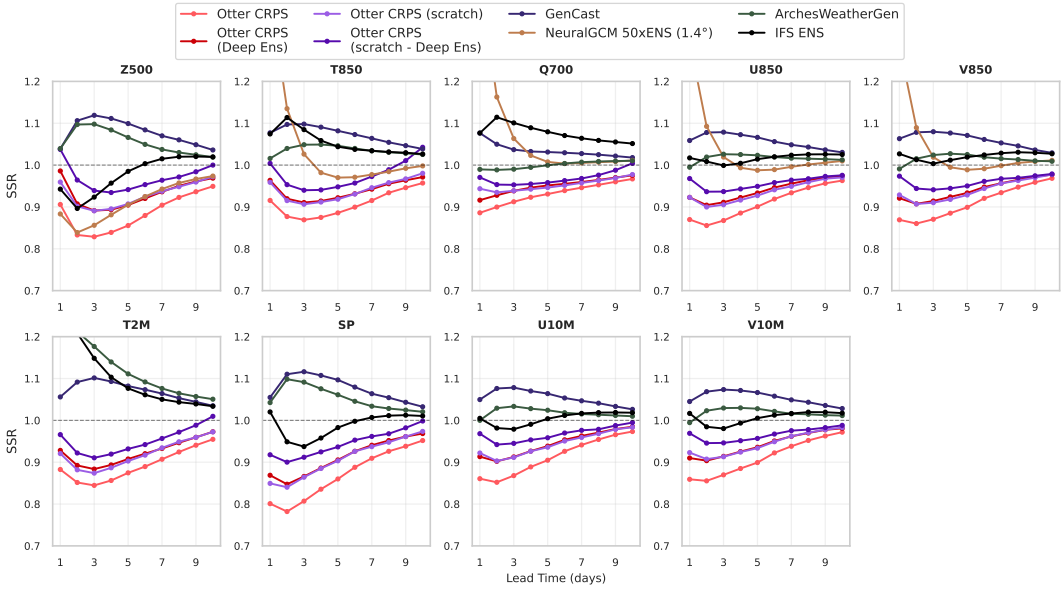


Figure 15: SSR over IFS ENS comparison for lead times up to 10 days for Otter CRPS, Otter CRPS (Deep Ens), Otter CPRS (scratch), Otter CRPS (scratch - DeepEns), GenCast, and ArchesWeatherGen, and NeuralGCM.

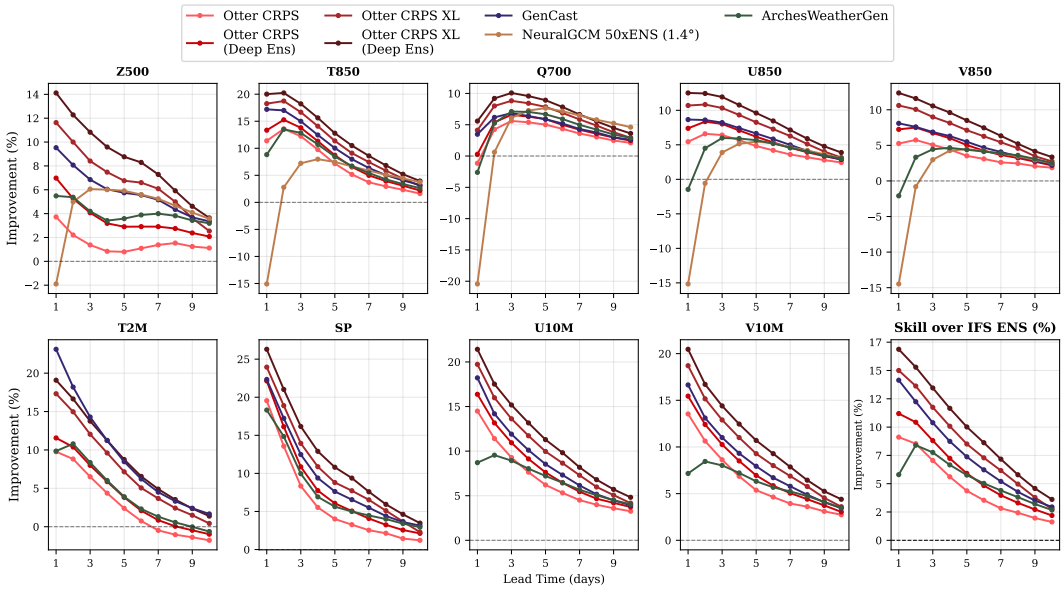


Figure 16: CRPS skill over IFS ENS comparison for lead times up to 10 days for Otter CRPS, Otter CRPS (Deep Ens), Otter CPRS XL, Otter CRPS XL (DeepEns), GenCast, and ArchesWeatherGen, and NeuralGCM. The last plot shows the percentage improvement over IFS ENS averaged over the headline variables.

C.6 Example Predictions

Figures 18 to 26 provide forecast visualizations for the headline variables (Z500, Q700, T850, U850, V850, T2m, SP, U10m, and V10m) across lead times of 24h, 3 days, and 10 days. We compare the deterministic Otter base model with our two probabilistic variants: Otter CRPS and Otter CRPS (scratch). For the probabilistic models, we visualize two individual ensemble members alongside the ensemble mean calculated over five samples.

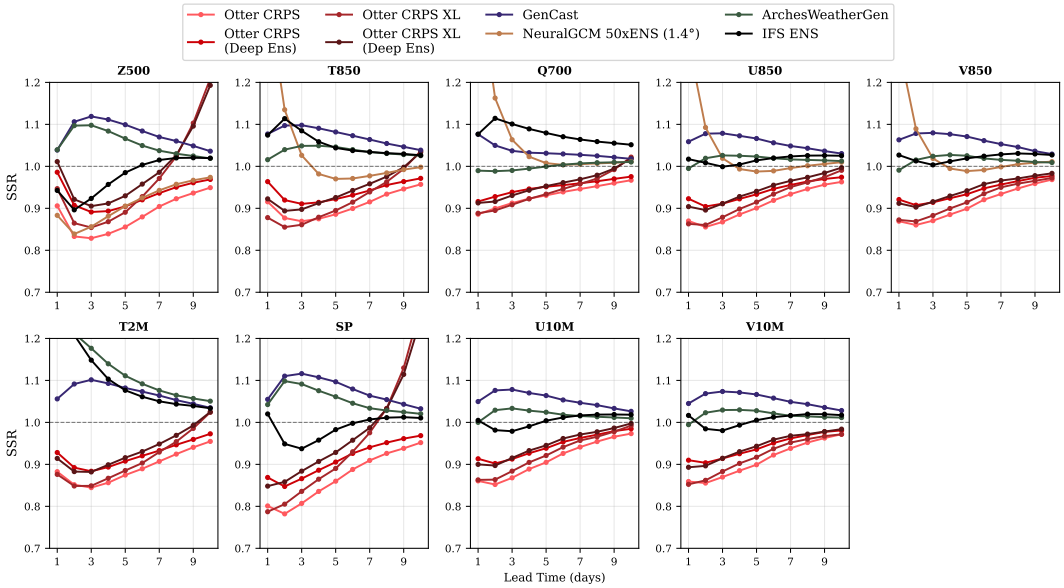


Figure 17: SSR over IFS ENS comparison for lead times up to 10 days for Otter CRPS, Otter CRPS (Deep Ens), Otter CPRS XL, Otter CRPS XL (DeepEns), GenCast, and ArchesWeatherGen, and NeuralGCM.

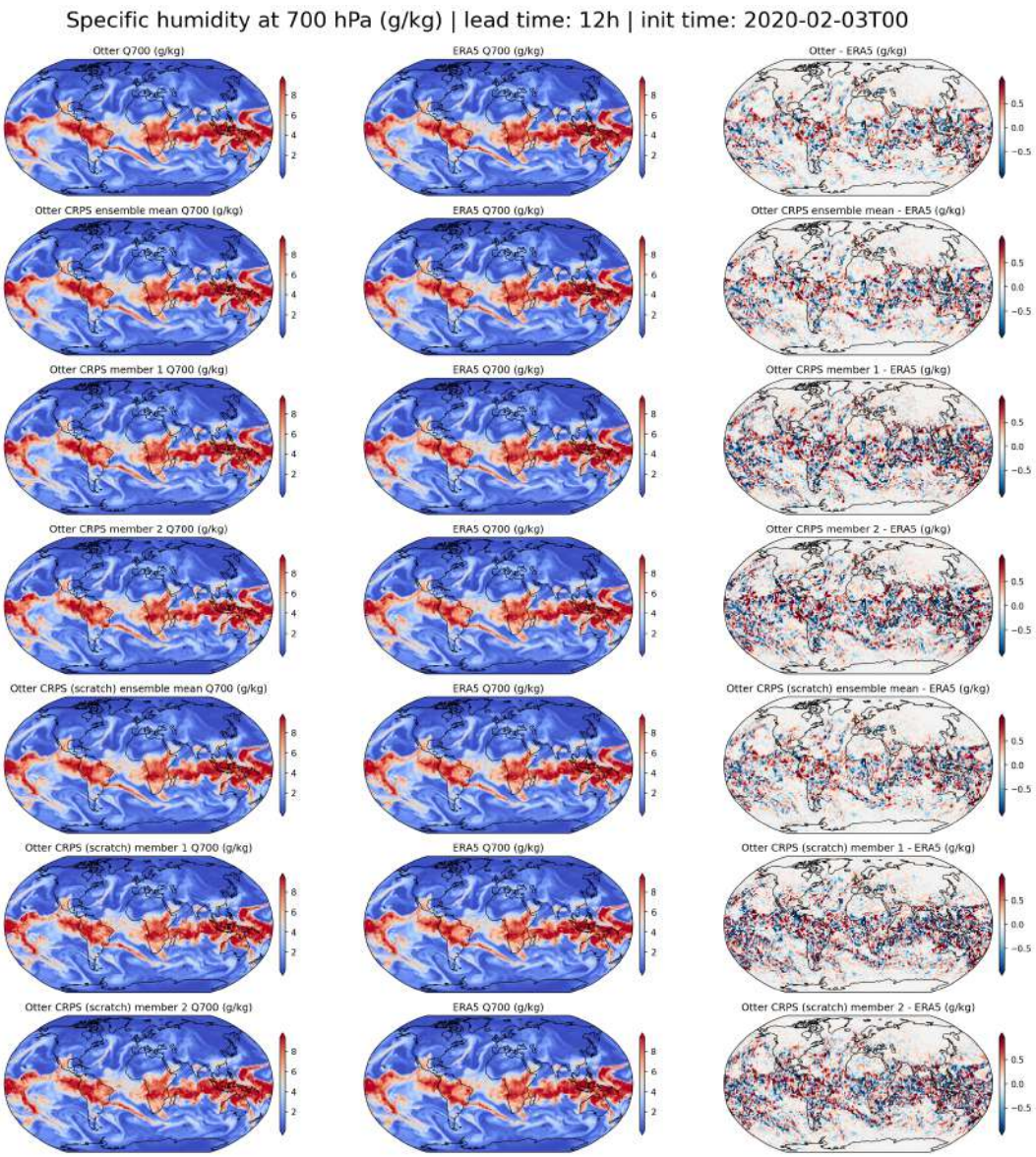


Figure 18: Visualization of Q700 at 12h lead time.

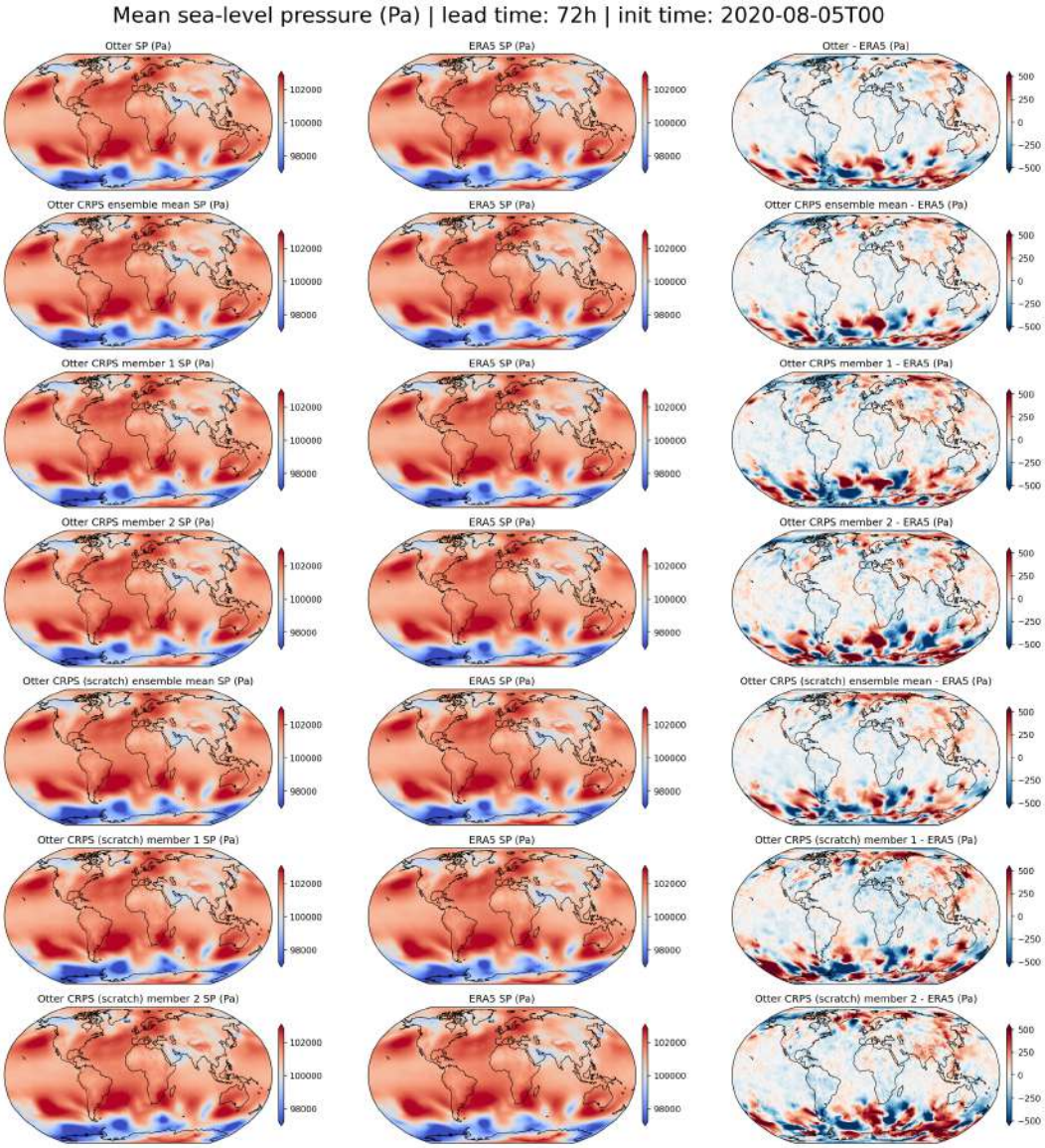


Figure 19: Visualization of SP at 3 days lead time.

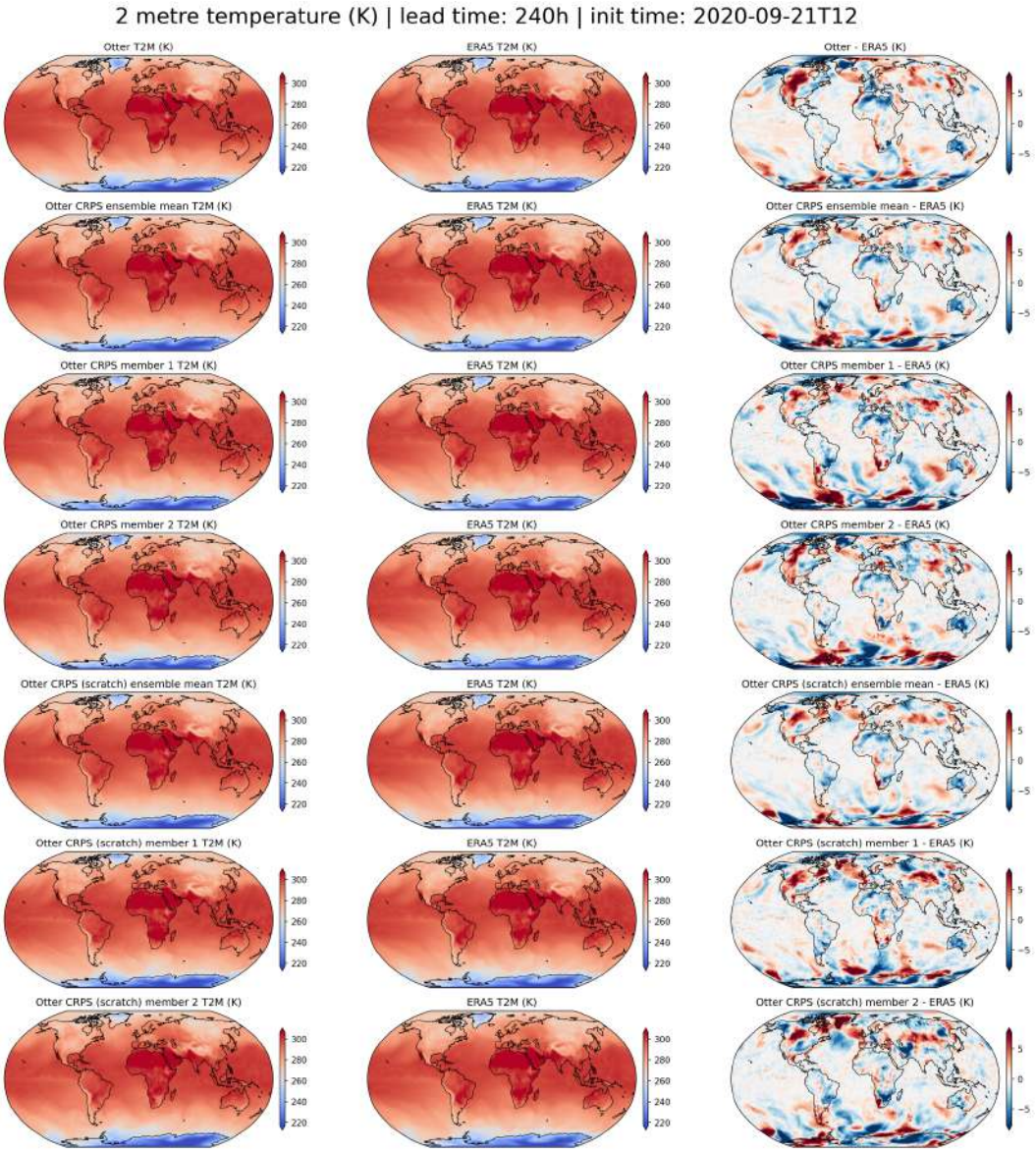


Figure 20: Visualization of T2M at 10 days lead time.

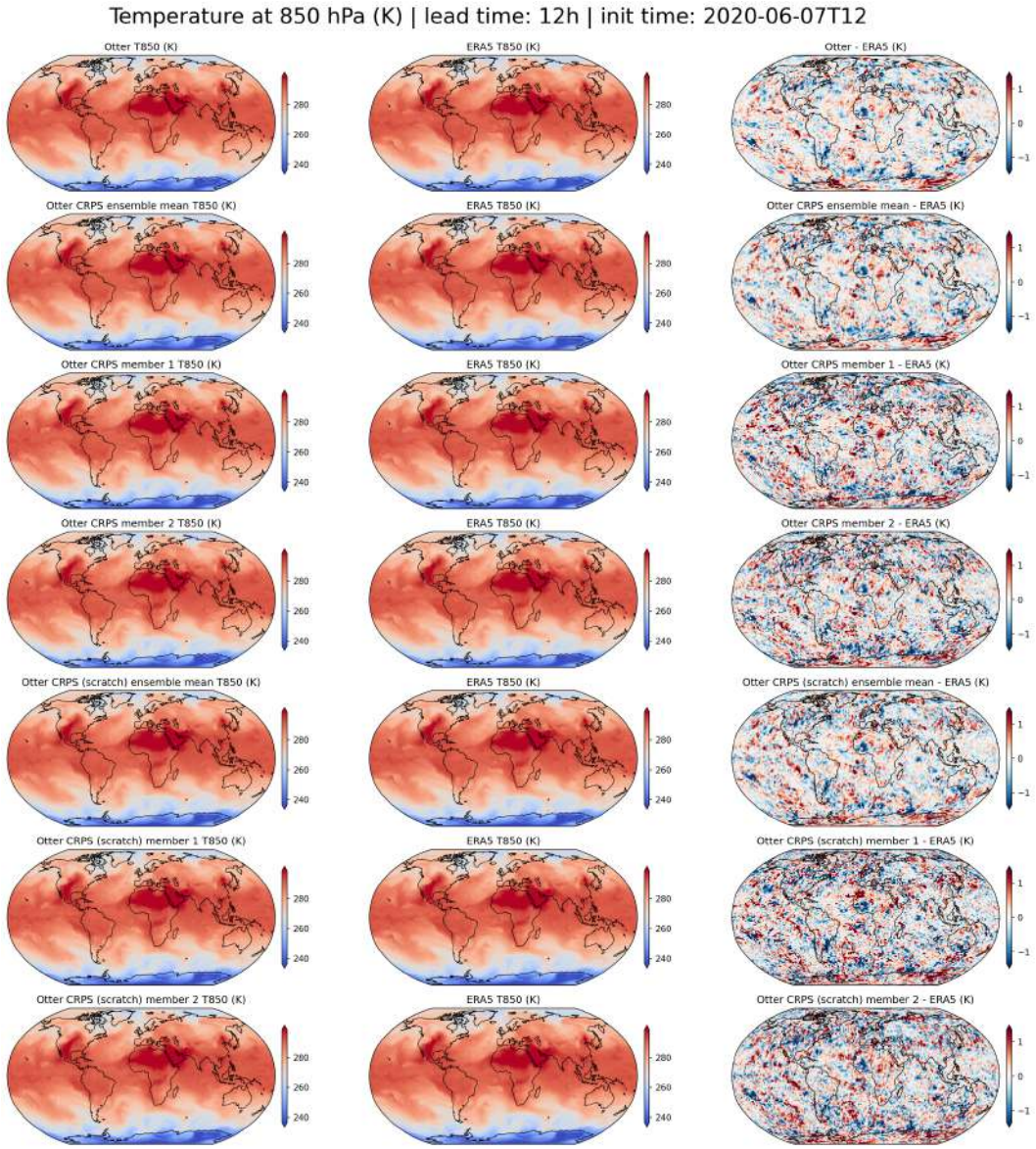


Figure 21: Visualization of T850 at 12h lead time.

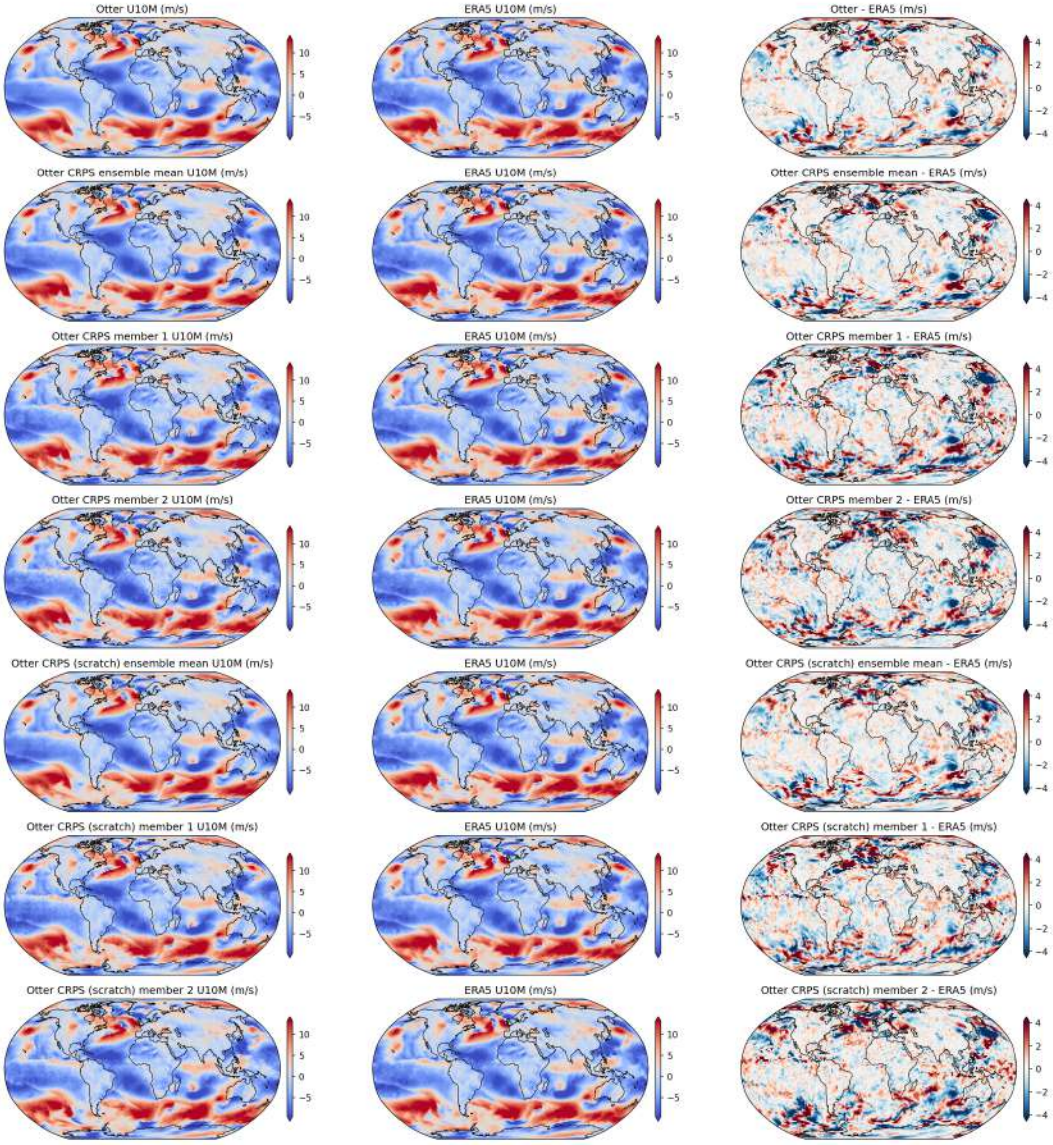


Figure 22: Visualization of U10m at 3 days lead time.

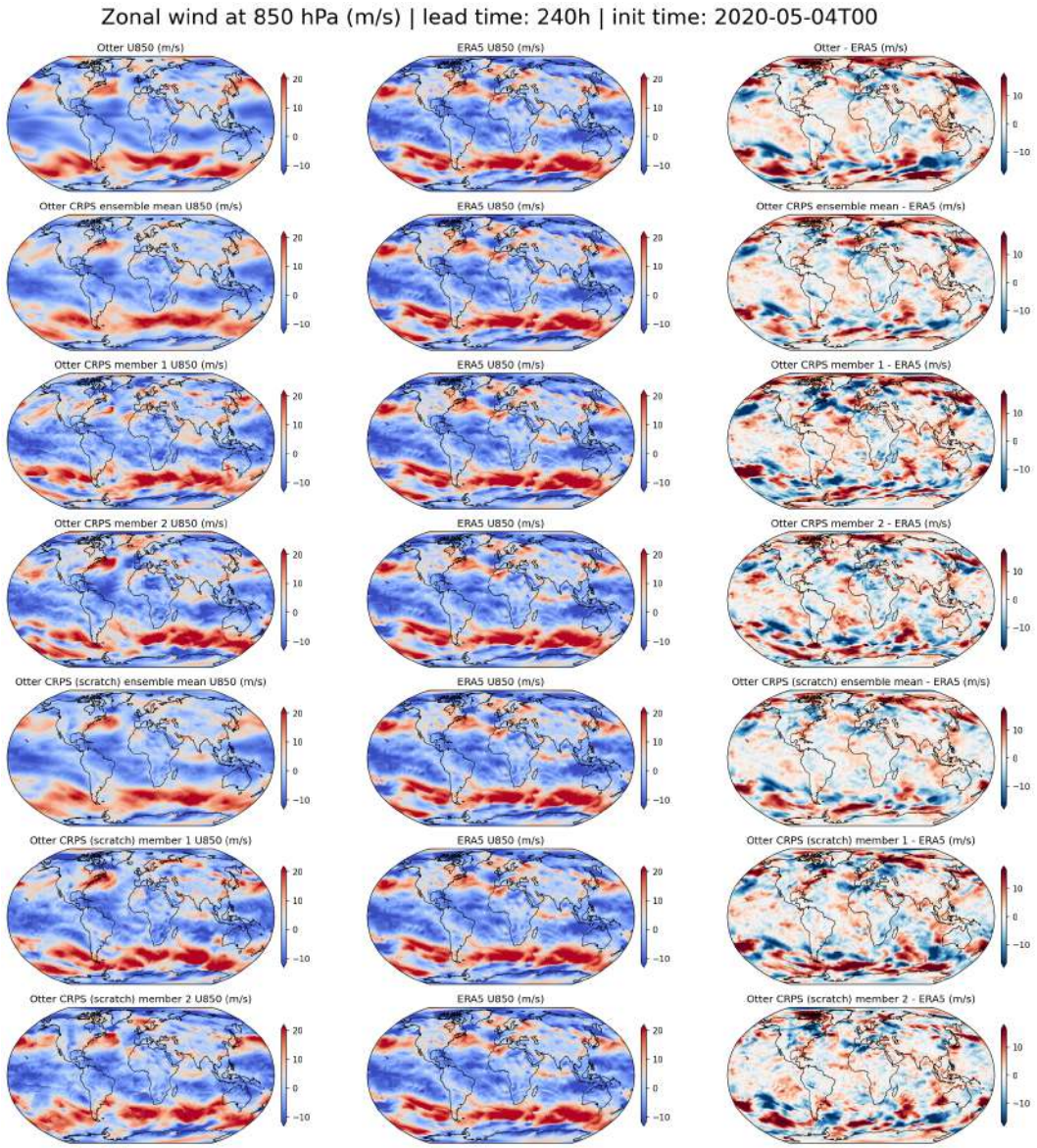


Figure 23: Visualization of U850 at 10 days lead time.

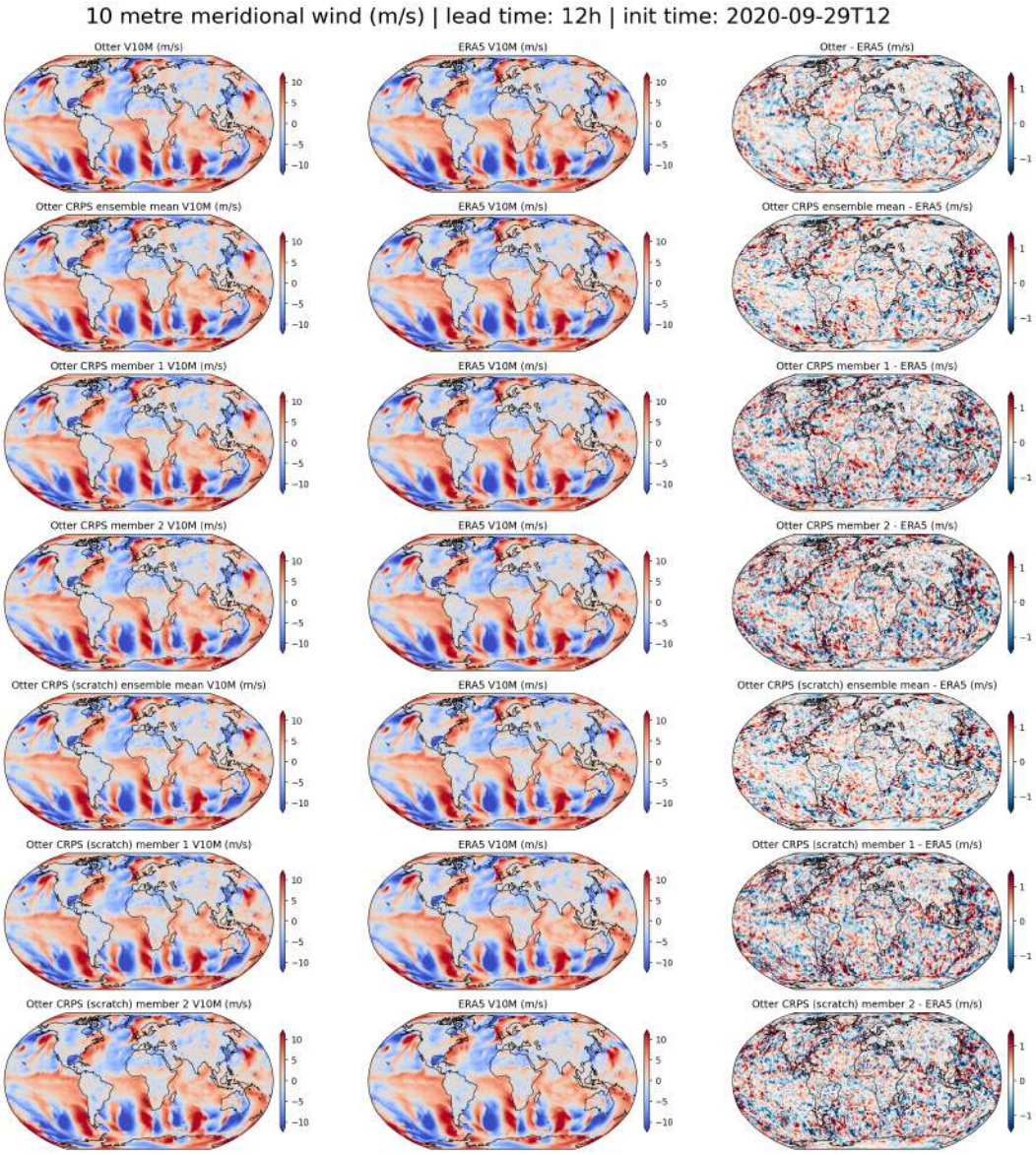


Figure 24: Visualization of V10m at 12 hours lead time.

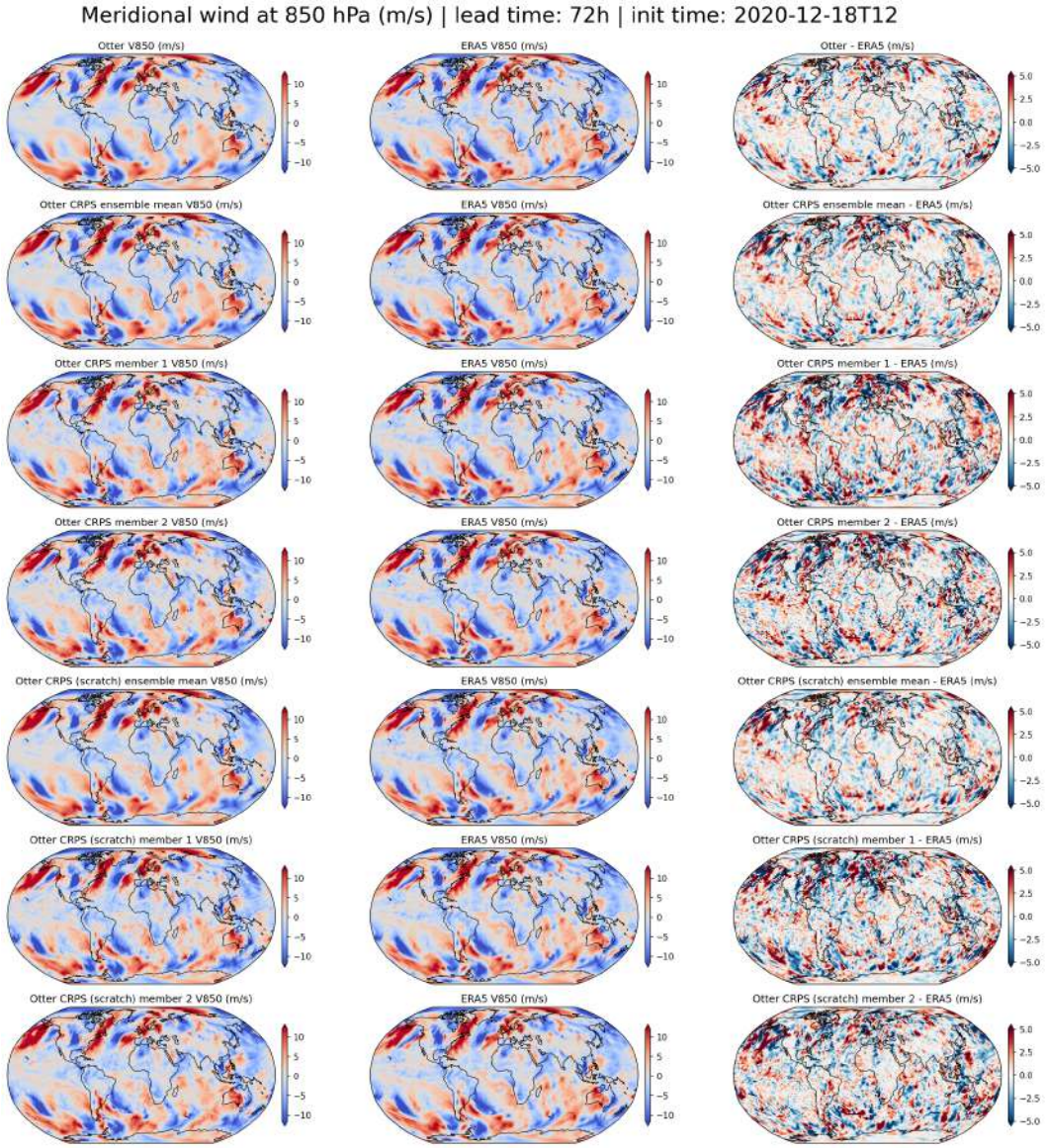


Figure 25: Visualization of V850 at 3 days lead time.

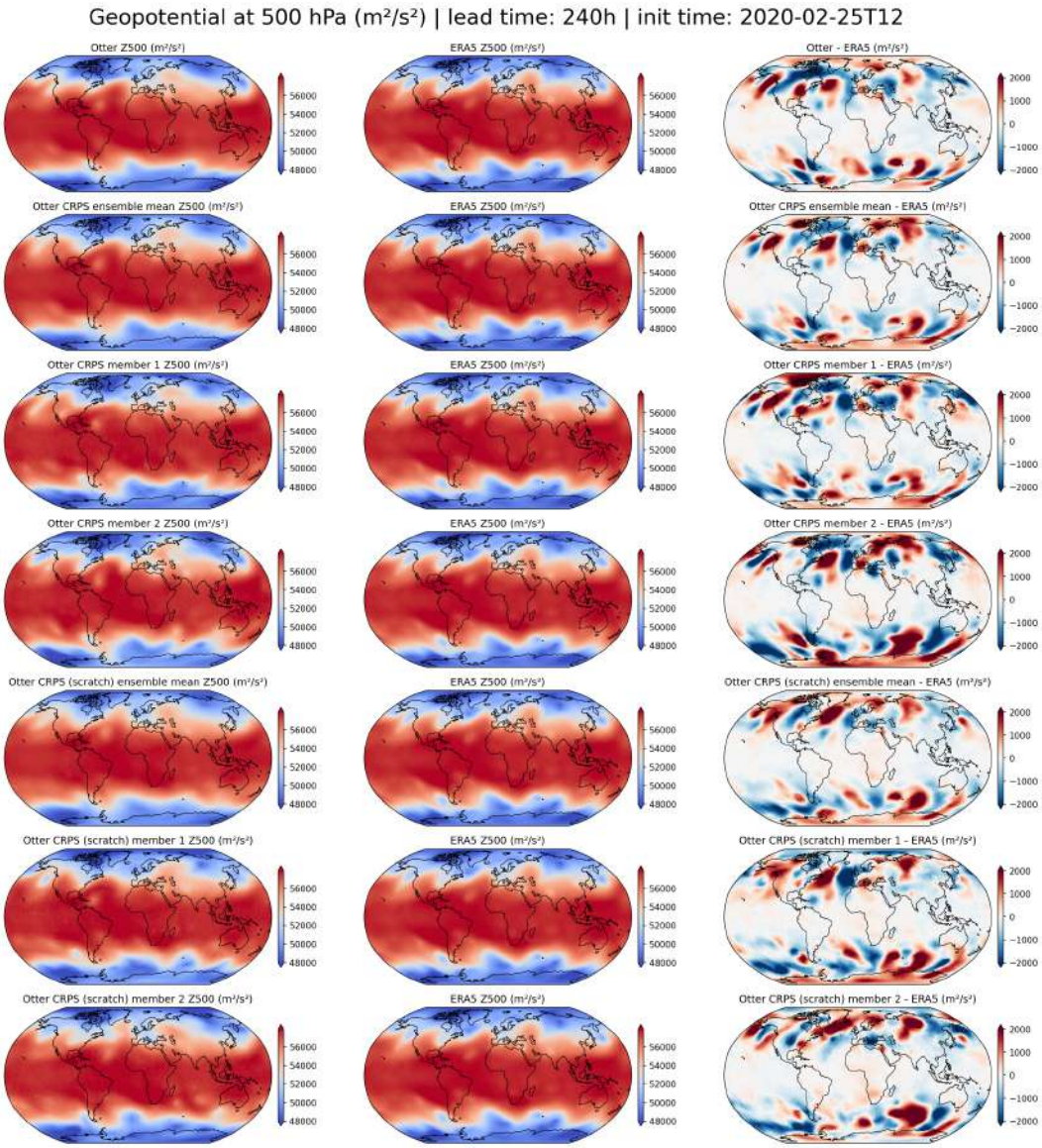


Figure 26: Visualization of Z500 at 10 days lead time.

D Additional Details on the PDE task

This section provides more details on the PDE modelling task on the acoustic scattering task from The Well [Ohana et al., 2024]. This dataset consists of simulations of acoustic equations that model the propagation of pressure waves through domains with varying scattering properties (in this case, random material inclusions). The data comprises 4,000 trajectories, with each trajectory containing 101 timesteps of 256×256 resolution images. It features multiple physical channels, including scalar fields for pressure, material density, and material speed of sound, alongside a vector field representing the velocity. The density and speed of sound fields are constant, and the model only predicts the pressure and velocity channels. For additional dataset characteristics, we refer to Ohana et al. [2024].

D.1 Evaluation Details.

For evaluation we report the one-step VRMSE evaluated on the test set specified by The Well. We additionally report the rollout VRMSE in fig. 27.

Following McCabe et al. [2026] we use the following definition for the VRMSE:

$$\text{VRMSE}(u, v) = \sqrt{\frac{\langle (u - v)^2 \rangle}{\langle (u - \langle u \rangle)^2 \rangle + \epsilon}}, \tag{6}$$

where $\langle \cdot \rangle$ denotes the spatial mean operator and we add $\epsilon = 10^{-6}$ for numerical stability.

D.2 Base Deterministic Configuration.

The base configuration for PDE task is:

- **Embedding Dimension:** $D = 384$;
- **Depth Profile (U-Net):** $[2, 8, 4]$ blocks per stage;
- **Attention Heads:** 16;
- **FFN Activation:** SwiGLU;
- **Positional Embeddings:** Absolute (Spatial) + Rotary (RoPE);
- **Regularisation:** Weight decay 0.1, Dropout 0.05, DropPath 0.05;
- **Optimiser:** Muon (momentum $\mu = 0.95$);
- **Learning Rate:** Cosine decay (max 2×10^{-4}) with a linear warmup of 500 steps;
- **Training Budget:** 10 epochs with a gradient accumulation factor of 4 (approx. $84,000$ optimisation steps). The equivalent Walrus budget (in terms of data samples) is about 17 epochs.

D.3 Normalisation Strategy.

Similarly to Walrus [McCabe et al., 2026], we use the Reversible Normalisation technique. However, unlike Walrus, we rescale the backbone prediction by the per-instance RMS, rather than the RMS of the residual.

Let the input to the network $X_{ctx} = X_{t-(L-1)\times\delta:t}$. The per-instance RMS is computed over the context window:

$$\sigma(X_{ctx}) = \sqrt{\frac{1}{|\Omega|\, L} \sum_{\tau=t-(L-1)}^{t} \sum_{\mathbf{x}\in\Omega} X_\tau(\mathbf{x})^2 + \varepsilon}. \tag{7}$$

The network update with instance normalisation becomes:

$$\hat{X}_{t+\Delta t} = X_t + \text{Backbone}\left(\frac{X_{ctx}}{\sigma(X_{ctx})}, \text{Emb}(t + \Delta t) \right) \cdot \sigma(X_{ctx}), \tag{8}$$

where the backbone operates on the normalised context and its output (an increment in normalised space) is rescaled back to physical units by $\sigma(X_{ctx})$ before being added to X_t.

D.4 Training Details.

We use the mean absolute error (MAE) as the training objective. All models are trained on the NVIDIA A100-PCIe (80GB) machines for 10 epochs. This is significantly less than the equivalent budget that Walrus allocates per-dataset, which amount to about 17 epochs for the acoustic scattering dataset. Each training round takes slightly less than 2 A100 days.

D.5 Additional Results.

In addition to the one-step VRMSE presented in fig. 6, we evaluate the architectural variants using rollout VRMSE. Following the protocol of McCabe et al. [2026], these rollouts are computed over the entire trajectory length, initialised at time step 17. Consistent with our one-step findings, the majority of ablations degrade rollout performance. There are, however, some exceptions: models utilising AdamW, or a backbone configuration of $D = 640$ with a [1, 4, 1] depth profile, achieve superior rollout metrics at a comparable computational cost. Despite these nuances, we base our primary evaluation on one-step VRMSE. Because the models are trained using a one-step objective, this metric serves as a more direct indicator of core architectural effectiveness. While long-horizon rollout stability could be addressed via techniques such as RFT, the relationship between base architecture and rollout performance remains an interesting area for future investigation.

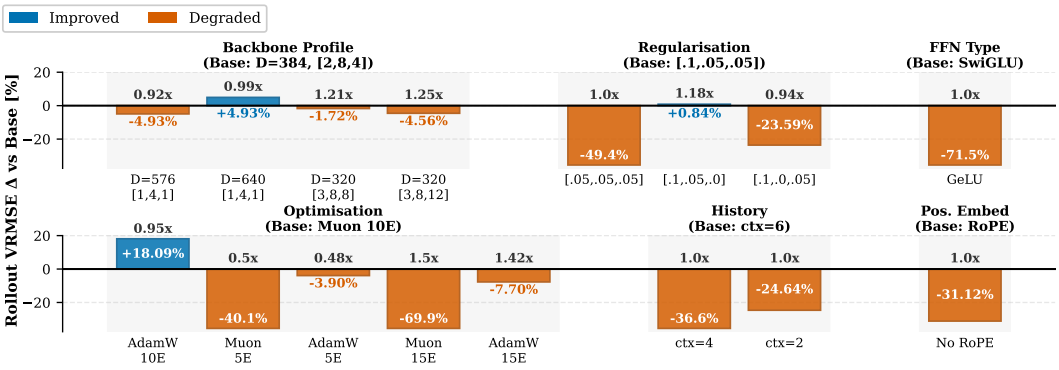


Figure 27: Ablation study on the acoustic scattering PDE task. We report the rollout VRMSE (over entire trajectory, initialised from time step 17, as performed in McCabe et al. [2026]) over the test set. The results confirm that findings from the weather domain (e.g., superiority of Muon, SwiGLU, RoPE, and balanced backbones) transfer to other physical systems.

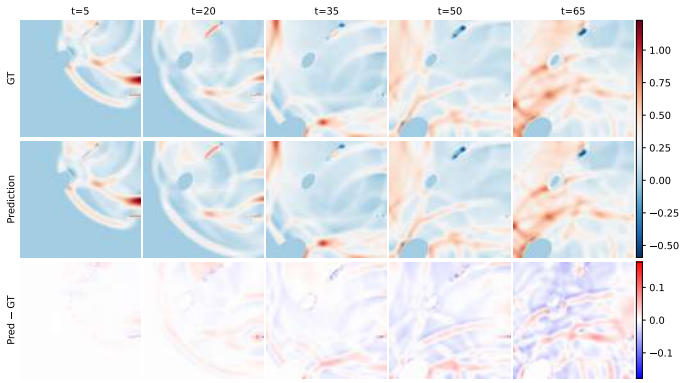
D.6 Example Predictions.

Figure 28 displays the ground truth, model predictions, and corresponding errors across five time steps of an example acoustic scattering trajectory. Results are shown for all three predicted channels: pressure and the vector velocity field.

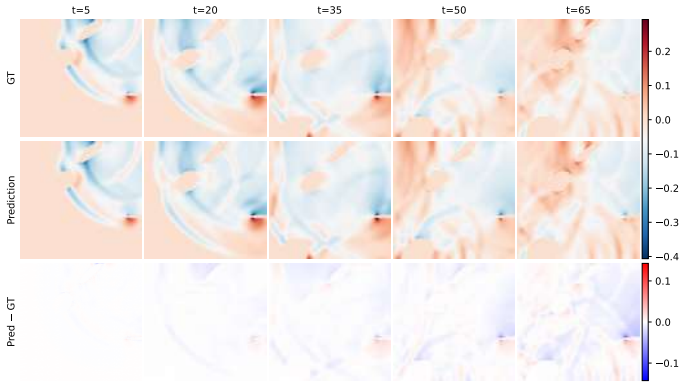
E Limitations and Future Work

Empirical Scope & Scaling. Our findings are empirical, resulting from a budget constrained to prioritise rapid iteration. Establishing formal scaling laws or theoretical guarantees remains future work. However, Otter Weather's efficiency makes it an ideal platform for previously prohibitive scaling studies across different resolutions, downstream tasks, and architectures.

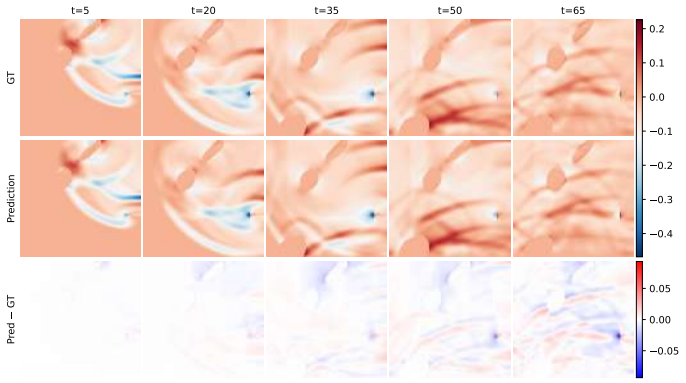
Optimal Branching for Deep Ensembles. While our results demonstrate that deep ensembling consistently yields significant improvements in probabilistic skill, the optimal methodology for constructing these ensembles remains an open question. Specifically, it is unclear at which stage of the training pipeline branching introduces the most beneficial, well-calibrated diversity. Future work should systematically investigate these trade-offs, particularly evaluating whether branching exclusively during the final RFT phase provides sufficient diversity compared to branching earlier, immediately following deterministic pre-training. Furthermore, further comparison is needed between



(a) Pressure channel.



(b) Velocity (x-axis) channel.



(c) Velocity (y-axis) channel.

Figure 28: Example ground truth (GT) (top, predictions, and errors (prediction - GT) for an acoustic scattering trajectory at 5 different time steps.

ensembles generated via different RFT rounds on a single from-scratch model and those composed of entirely independent from-scratch training runs. A rigorous analysis of how these varying branching points impact the balance between computational cost, ensemble spread, and predictive skill will be crucial for establishing best practices in efficient probabilistic modelling.

Spatial Resolution Comparisons. We train at a 1.5° resolution, whereas some baselines like GraphCast operate at 0.25°. Because the majority of forecasting skill on test data derives from lower-frequency spatial components, this higher resolution likely contributes minimally to overall skill on the metrics we consider while significantly inflating compute constraints. While evaluating a

1.5° variant of GraphCast would provide a more direct comparison, the computational cost of the 0.25° variant is $134\times$ higher than Otter Weather's; even at a reduced resolution, its compute footprint would likely remain substantially larger. Furthermore, GNN-based architectures are notoriously difficult to implement and train from scratch, and they lack the highly optimised hardware support of Transformer-based models. Consequently, we only evaluate against the official open-source release of GraphCast operating at 0.25°.

Inductive Biases. By prioritising general-purpose ML components, we trade some explicit physics-based inductive biases for simplicity. However, this does not preclude the value of domain-specific modifications; on the contrary, many of our findings are orthogonal to such techniques, and may lead to compounding benefits—a direction we leave for future investigation.

Broader Societal Impacts. While Otter Weather significantly lowers the computational barrier to entry for global weather modelling, this democratisation carries inherent risks. By making it feasible for practitioners without extensive meteorological or high-performance computing backgrounds to train and deploy predictive models, there is a risk of these systems being utilised in safety-critical domains (e.g., disaster management, agriculture, or aviation) without proper expert oversight. Furthermore, because we prioritise general-purpose ML components over strict physics-based inductive biases, the model may occasionally produce unphysical forecasts or miscalibrated uncertainty estimates. Over-reliance on such outputs by non-experts could lead to adverse real-world decisions. Finally, while our architecture is highly computationally efficient, the development and training of deep learning models still contribute to carbon emissions, underscoring the need to balance rapid iteration with environmental responsibility.